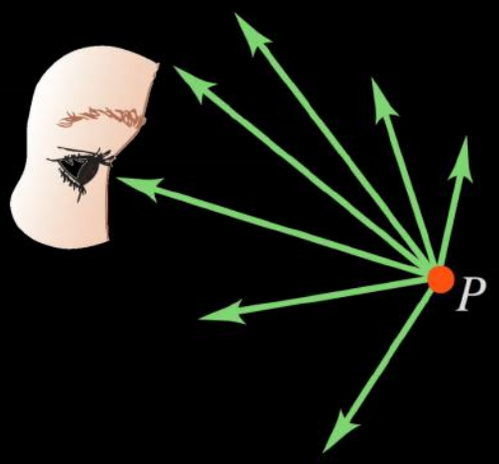
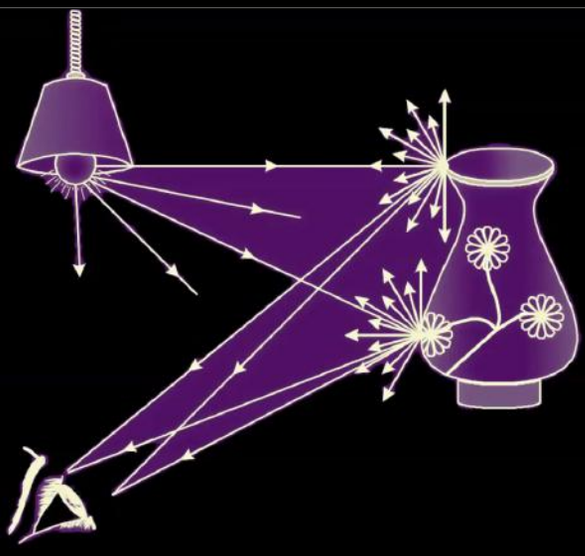
Rectilinear Propagation Of Light:



Light travels in straight line unless its medium of propagation changes on it is made to pass through small apertures of the size comparable to its wavelength. (Change of medium leads to refraction and passing through small apertures leads to diffraction about which we will study in wave optics). Whenever light falls on any object, a portion of it is reflected in various directions. When these reflected rays enter our eyes, an image is formed on the retina of our eyes. Images are formed at the intersection point of rays emanating from any point. Typically a cone (or pencil) of rays from any point fall at the pupil of our eye which then get processed by our eye lens and finally the image is formed at our retina.

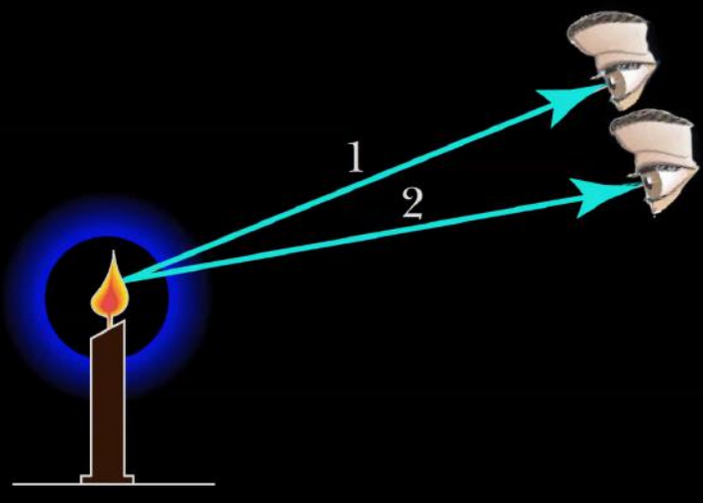
This process is schematically illustrated in adjacent figure. Each particle on surface of the flower Vase sends out a pencil of rays, which converge at different points of retina. The light sensors on the retina convert this data in to electrical signals and the brain the synthesizes the complete image.



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Parallax And Depth Perception:

Whenever we view any object with both eyes, each eye makes its own image, thus these two images are slightly shifted from each other because of rays coming at different angles for both eyes. Nearer the object, greater the shift in this image. This shifting of image



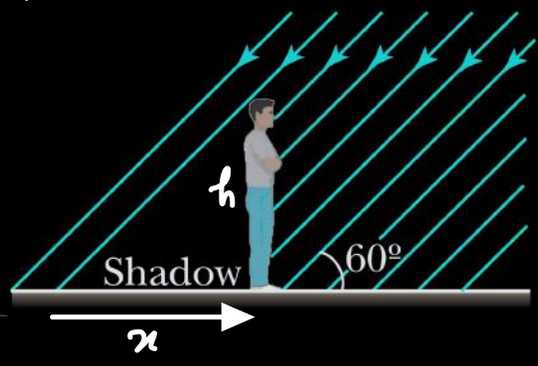
upon looking from each eye individually is known as parallax. For farther objects, this parallax is smaller. (Try to look at your pen holding it at various distances and viewing it with only one eye at a time). Our brain processes these two slightly different images and based on parallax, we are able to estimate the depth of the objects.

SHADOW FORMATION: Whenever light rays are obstructed by an opaque object we get a region of space with absence of light which is also known as shadow region.

EXAMPLE:

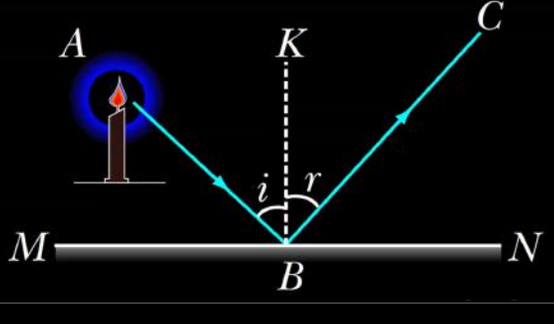
Assume that sun rises at 6:00 A.M. and sets at 6:00 P.M. Find the angle made by the sunrays with the horizontal at 10:00 A.M. Also find the length of the shadow of a man 6 ft high at this time. Assume that the man is in the plane in which sun appears to move.

Solution: From eastern horizon to western horizon, the sun covers an angle of 180° in a span of 12 hours. Thus, in 4 hours, it must appear 60° above the horizon with rays coming as shown in the figure. Now from the figure, $h/x = \tan 60^\circ \Rightarrow$ length of shadow, $x = h/\tan 60^\circ$ i.e $6/\sqrt{3}$ ft. or length of shadow: $2\sqrt{3}$ ft Ans.



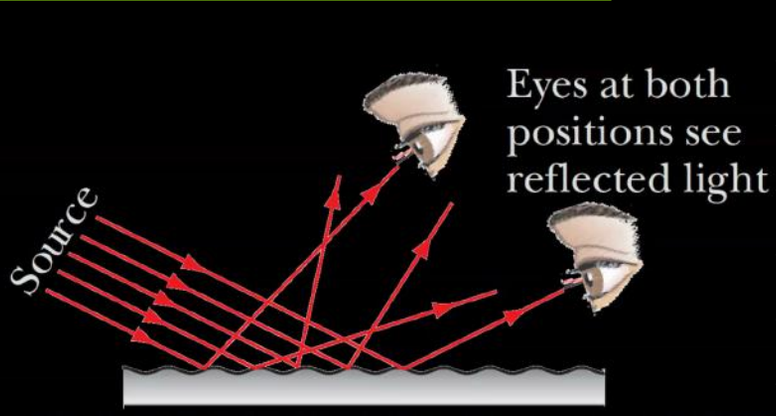
LAWS OF REFLECTION:

1. The angle of incidence is equal to angle of reflection
2. The incident ray, normal and reflected ray lie in the same plane.

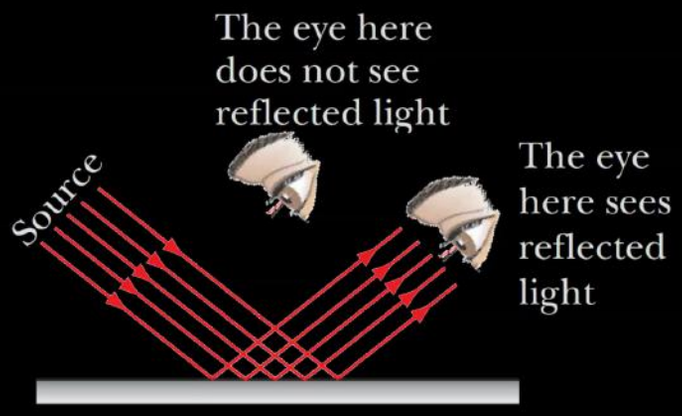


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Specular Vs Diffuse Reflection:



Diffuse Reflection (Rough surfaces)



Specular Reflection (Smooth Finish).

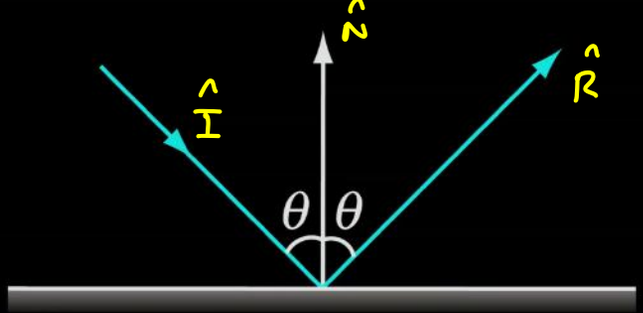
When light coming from a particular direction fall on a rough (or Matte finish) surface, it gets reflected in various directions due to surface roughness. This type of reflection is called diffuse reflection. However when light from a particular direction falls on a smooth polished surface, it gets reflected only in a particular direction. This type of reflection is called specular reflection.

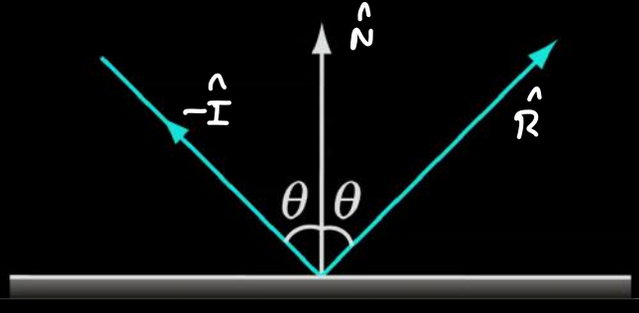
Vector Form Of Laws Of Reflection:

Let $\hat{I}$, $\hat{N}$ and $\hat{R}$ be the unit vectors along incident ray, Normal and reflected rays respectively. Recognise that the vectors $-\hat{I}$ and $\hat{R}$ are equally inclined to $\hat{N}$. Thus $-\hat{I} + \hat{R} = 2\cos\theta\,\hat{N}$, also Since $\hat{I}$ & $\hat{N}$ are unit vectors thus $-\hat{I}\cdot\hat{N} = \cos\theta$, thus:

$$-\hat{I} + \hat{R} = 2(-\hat{I}\cdot\hat{N})\hat{N}$$

$$\Rightarrow \boxed{\hat{R} = \hat{I} - 2(\hat{I}\cdot\hat{N})\hat{N}}$$



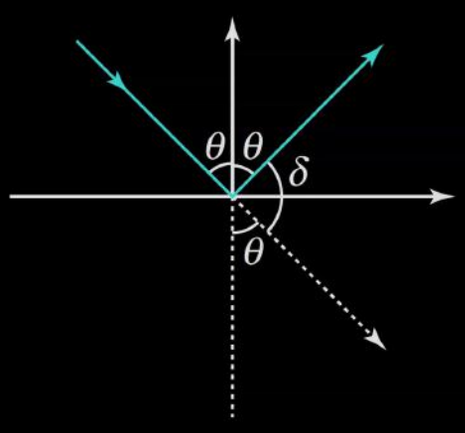


→ This formula can be conveniently used to find reflected ray for arbitrary orientations of the mirror specified in terms of $\hat{i}$, $\hat{j}$ and $\hat{k}$ vectors.

→ Upon reflection from a plane mirror, component of unit vector ⊥ to the mirror gets reversed and the other component remains unchanged.

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Angle Of Deviation From A Mirror:



Angle between the incident ray and reflected ray is known as angle of deviation. Consider the adjacent figure. From this we can clearly see that Angle of deviation:

$$\delta = \pi - 2\theta$$

Here θ is the angle of incidence.

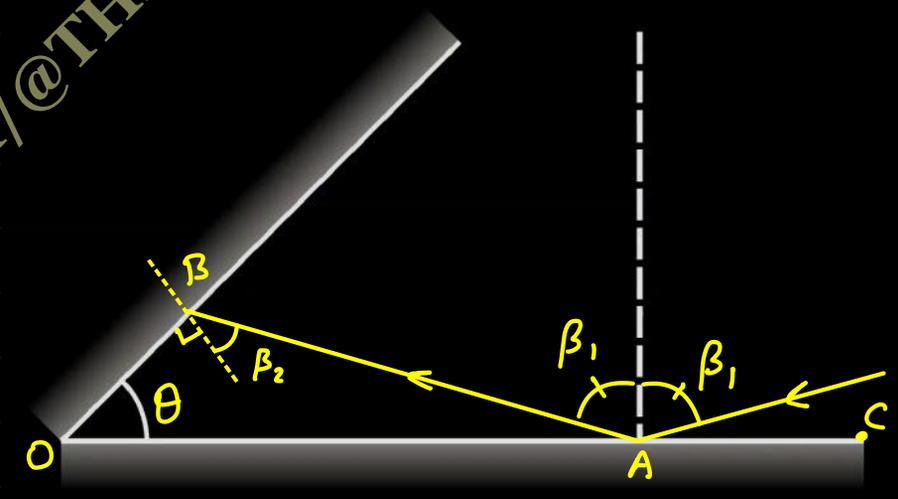
→ If the mirror's normal is rotated through an angle ϕ clockwise, reflected ray is rotated through an angle 2ϕ clockwise.

→ If incident ray is rotated by an angle α clockwise reflected ray rotates by an angle α anticlockwise.

→ If both the incident ray and normal are rotated, net effect on reflected ray is found by superposition. (i.e. add the two effects considering one at a time).

Incidence On System Of Pair Of Mirrors At An Angle:

Whenever we have a system of mirrors, successive angles of incidence decrease in AP, i.e $\beta_2 = \beta_1 - \theta$, $\beta_3 = \beta_2 - \theta$ etc.



This can be proved as follows

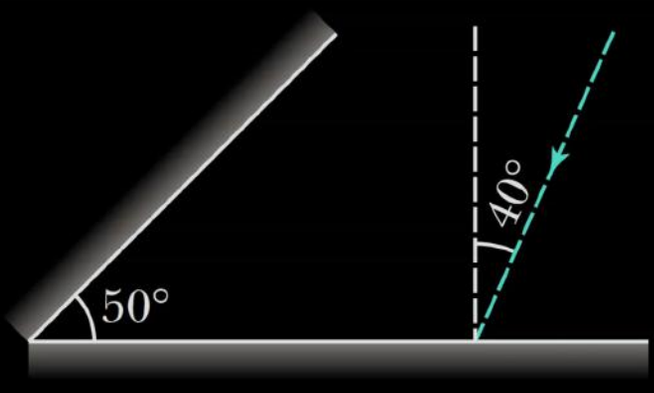
$\angle BAC = \angle OBA + \angle BOA$

$\Rightarrow 2\beta_1 + (90^\circ - \beta_1) = 90^\circ + \beta_2 + \theta$

$\Rightarrow \beta_2 = \beta_1 - \theta$, thus at n^{th} reflection, $\beta_n = \beta_1 - (n-1)\theta$. A negative angle of incidence means that the ray was moving away from O while approaching the point of incidence and a positive angle of incidence means that the incident ray was moving inwards (means to O). To find the total number of reflections, we can keep generating the terms of the A.P. $\beta_n = \beta_1 - (n-1)\theta$ until $|\beta_n|$ remains less than 90° (∵ an incidence angle of -90° means ray is parallel to the mirror).

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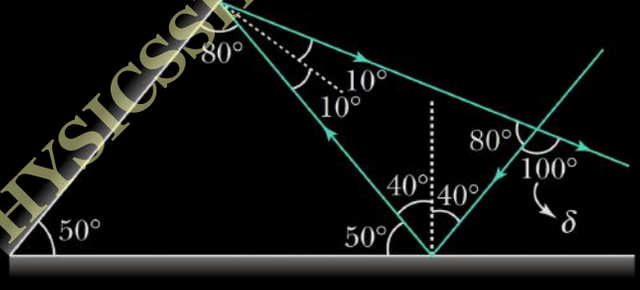
EXAMPLE: For the shown incident ray, find (a) Angles of incidence at successive reflections until there are no more reflections (b) Total number of reflections (c) Deviation angle of the ray after two reflections. (Mirrors are infinite).



Solution: (a) $\beta_1 = +40°$, $\beta_2 = \beta_1 - \theta = -10°$

$\beta_3 = -10 - 50 = -60°$ after this no more reflections are possible $\because -60 - 50 = -110°$ which is less than $-90°$. Thus required incidence angles are $40°$, $-10°$ and $-60°$ (b) Total 3 reflections

(c) Consider the ray diagram shown in adjacent figure. Deviation at first mirror: $\delta_1 = 180° - 2\times 40°$ i.e $100°$ clockwise. Similarly at the second mirror, $\delta_2 = 180° - 2\times 10°$ $\Rightarrow \delta_2 = 160°$ clockwise, thus total deviation $\delta = \delta_1 + \delta_2 = 100° + 160° = 260°$ c.w.



Now Since $360°$ is a full circle, $260°$ c.w. is equivalent to $100°$ anticlockwise (Notice that successive clockwise deviations are, $180° - 2\beta_1$, $180° + 2\beta_2$, $180° - 2\beta_3$ and so on with β_i being signed values)

→ Think: what will be the cumulative deviation after 3rd reflection?

Solution: $\delta = \delta_1 + \delta_2 + \delta_3 = (180° - 2\times 40°) + [180° + 2(-10°)] + (180° - 2(-60°))$

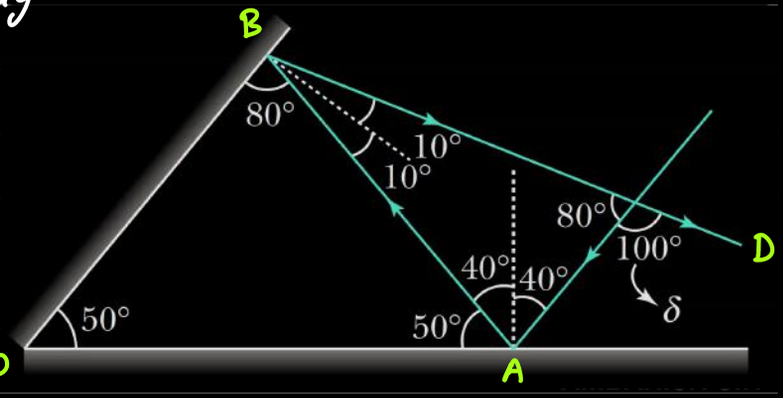
$= 100° + 160° + 300° = 560°$ clockwise

Now $720°$ clockwise would make a perfect circle and $560°$ is $160°$ short of $720°$ thus effective deviation will be $160°$ anticlockwise.

EXAMPLE: Through what angle will the ray BD rotate and in which sense if:

(a) lower mirror is rotated by $1°$ anticlockwise

(b) lower mirror is rotated $1°$ anticlockwise and upper mirror is rotated $1°$ clockwise.



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Solution: We shall use superposition step by step.

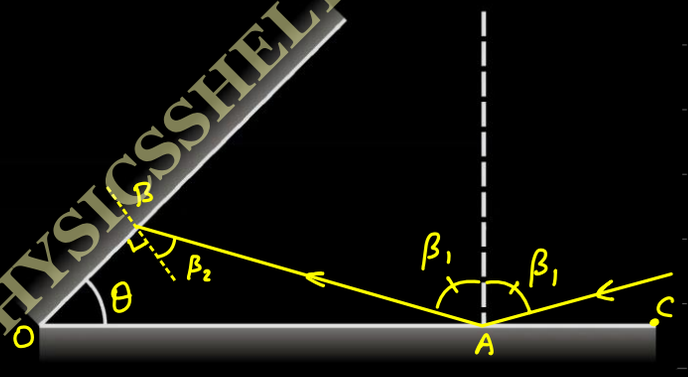
(a) Due to lower mirror rotating 1° anticlockwise, AB rotates 2° anticlockwise ⇒ reflected ray BD rotates 2° clockwise.

(b) If additionally mirror OB is rotated clockwise by 1°, it will cause an additional rotation of BD 2° clockwise thus by superposition, Ray BD will turn 4° clockwise.

Think: What would have happened if instead of clockwise, OB were also rotated anticlockwise?

Solution: In this case additional rotation would have been 2° anticlockwise thus net additional rotation of ray BD would be 2° − 2° i.e 0°.

EXAMPLE: Two mirrors are arranged at an angle θ as shown. Find the total deviation of the incident ray after two reflections.



Solution: $\delta = 180° - 2\beta_1 - (180° - 2\beta_2)$

i.e $\delta = 2\beta_2 - 2\beta_1 = 2(\beta_1 - \theta) - 2\beta_1$

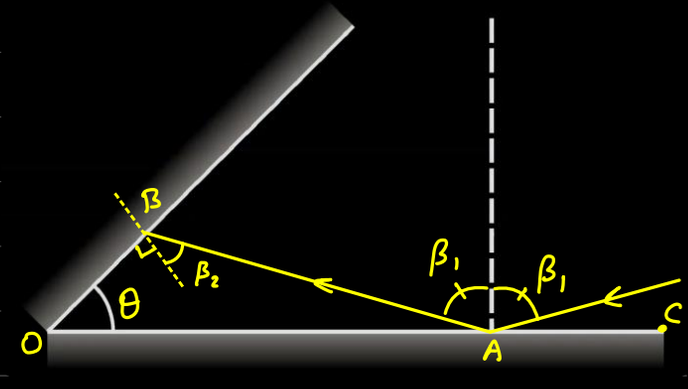
$\Rightarrow \quad \delta = -2\theta$ clockwise or 2θ anticlockwise.

→ Note that total deviation is independent of β_1, it only depends on θ. (This will also be the case with even number of reflections).

→ If $\theta = 90°$, the rays will make a full U turn (i.e 180° turn).

RETRACING OF PATH: If any ray falls normally on a mirror, it begins to retrace its path. (Normal incident ≡ angle of incident = 0°).

EXAMPLE: In the shown system of mirrors, $\beta_1 = 75°$, find the value of θ if upon 3rd reflection, the ray begins to retrace its path.

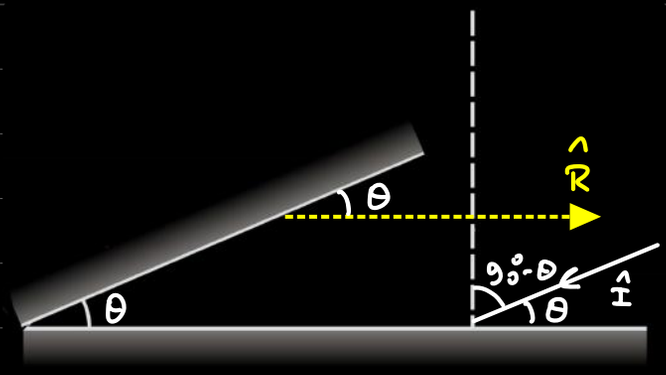


Solution: $\beta_3 = \beta_1 - 2\theta = 0°$ (∵ Normal incidence)

Thus $\theta = \beta_1/2 = 37.5°$ Ans.

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EXAMPLE: Find the angle θ between the mirrors if a ray incident parallel to one of the mirrors emerges parallel to the other mirror after four reflections.



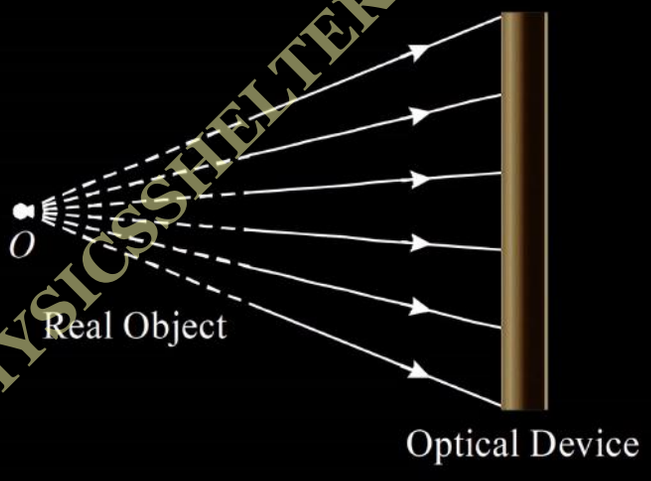
Solution: $\beta_4 = \beta_1 - 3\theta$

$\Rightarrow -(90^\circ - \theta) = (90^\circ - \theta) - 3\theta = 90^\circ - 4\theta$

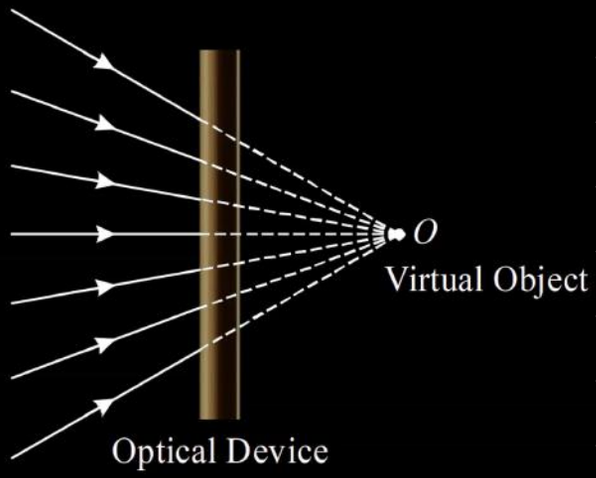
$\Rightarrow \quad 5\theta = 180^\circ \Rightarrow \theta = 36^\circ$ Ans.

REAL AND VIRTUAL OBJECTS:

When divergent rays coming from a point are incident on a reflecton or refractons, the point is said to be a real object. Note that the point might not be a real material particle, it could also be an intermediate image formed due to prior processing of rays by some optical devices.



When convergent rays are incident on an optical device then point of convergence of such rays is called virtual object for the optical device. The rays may not finally actually converge at this point due to being processed by the optical device, however if the device were not there then the convergent rays would actually intersect at this point.

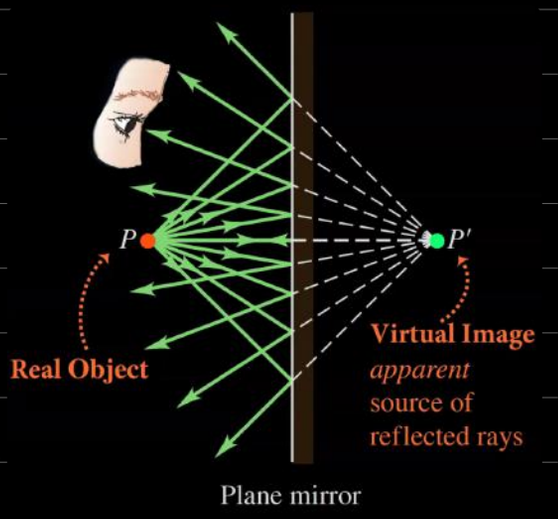


→ In case of parallel rays, if the source is a material object such as the sun or a star, we call it a real object, otherwise we may call it either virtual or real object.

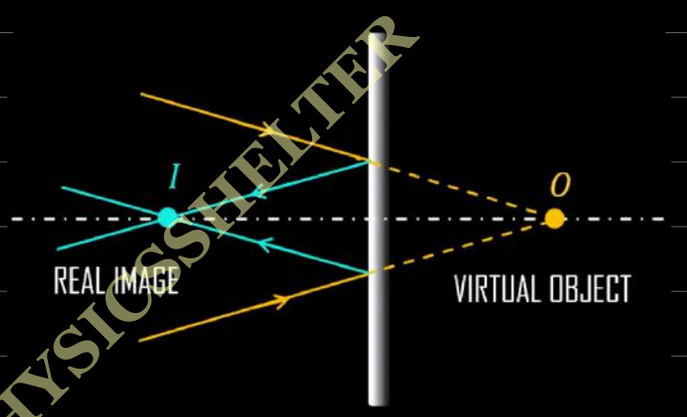
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Real And Virtual Images:

If upon being processed by the optical device, divergent rays emerge, then point point from which divergent rays appear to emerge is called the virtual image. We may not be able to obtain a virtual image on screen, however we can see it and photograph it just like we can photograph any real object.



When convergent rays emerge from an optical device then the point of convergence of these emergent (or reflected) rays is called the real image. A real image can be obtained on a screen suitably placed at the plane of convergence of rays.



→ Image is formed at intersection point of emergent/reflected rays.

Principle Of Optical Reversibility:

If a light ray is reversed (or we imagine another light ray traveling in reverse direction), it completely retraces its path. This essentially implies that object and image can be swapped and it will still lead to a valid ray diagram according to Geometrical optics. (For instance consider swapping O and I in the above ray diagram with the lens arrangement).

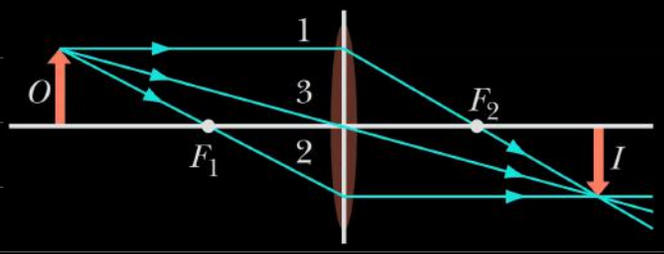
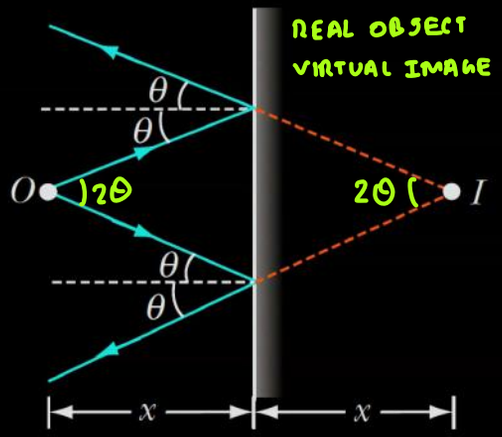
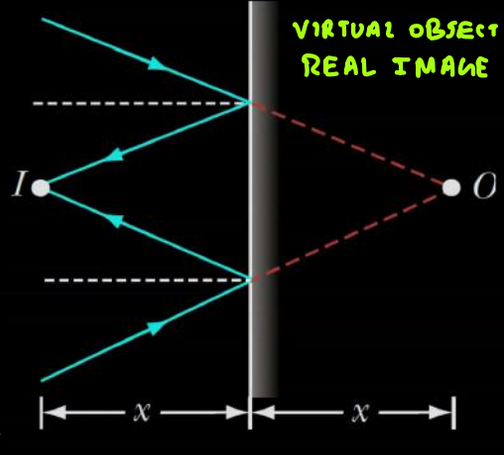


Image Formation From A Plane Mirror:



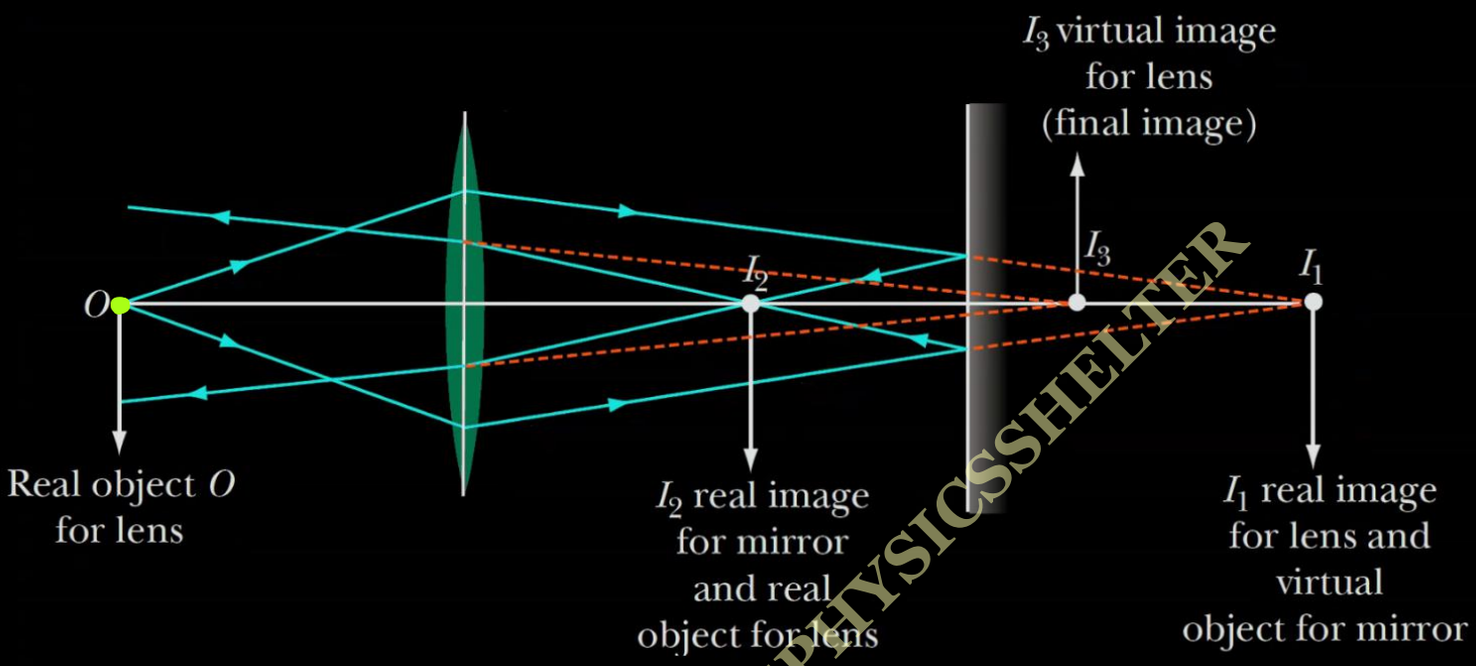


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For locating an image we need an intersection of two rays. Consider the above ray diagrams for image formation from a plane mirror.

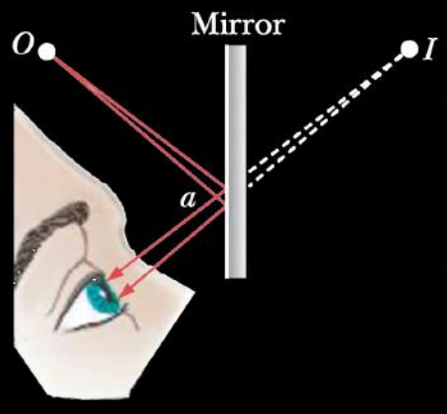
→ Image and object are equidistant from the mirror.

→ For real object image is virtual and for virtual object, the image is real.

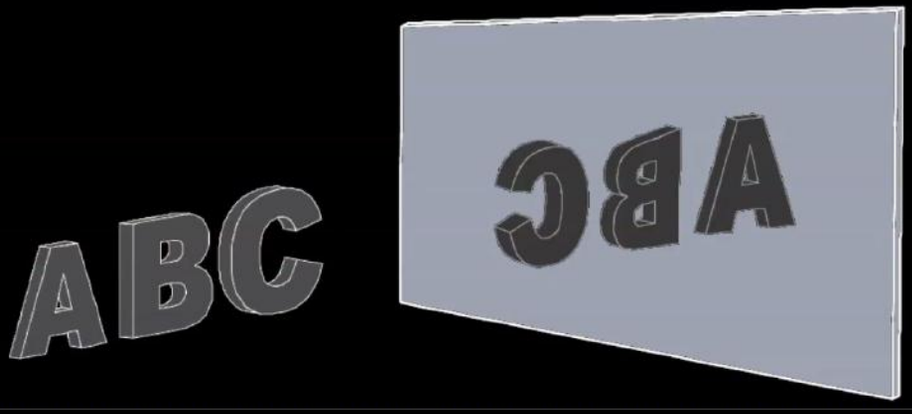


→ A virtual object can be generated using a converging lens (for instance see the above figure

→ A pencil of rays from O enters the eye after reflection from the mirror. Only a small portion of mirror near a is involved in this reflection. The light appears to originate from point I behind the mirror and hence we see an image there.



→ Image formed from a plane mirror is erect and longitudinally inverted (see figure).



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→ Not all the rays reflected from a mirror will enter an observer's eye from every point of view, thus thus the portion of image visible from a point of view depends on the point of view itself.

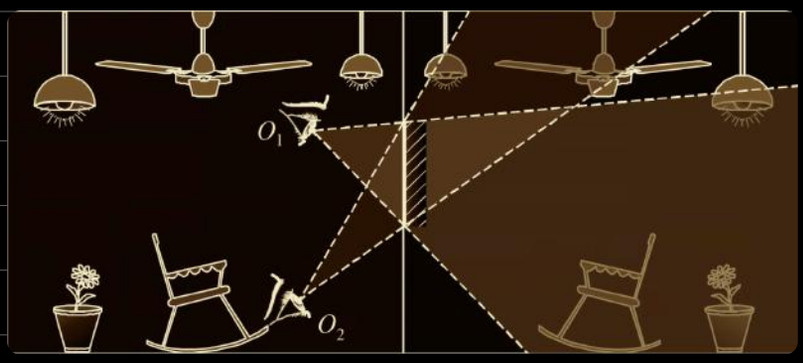
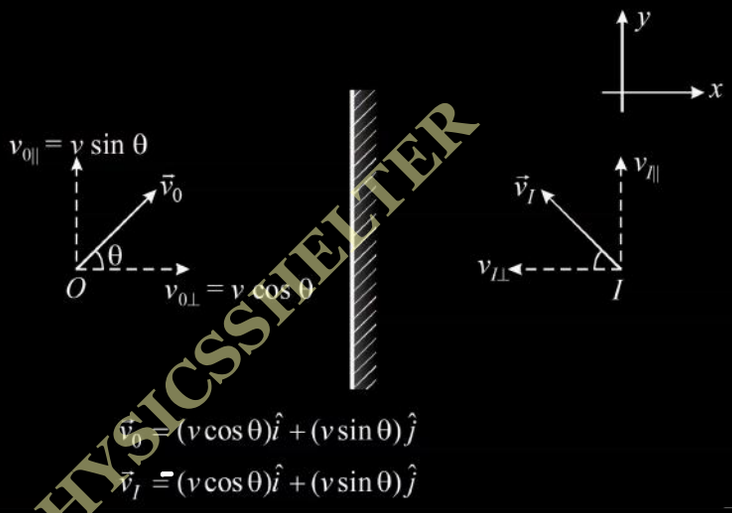


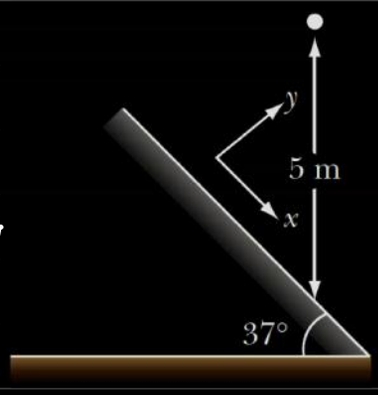
IMAGE VELOCITY:

Recall that in a previous section we saw that perpendicular distance of object from mirror is same as that of the image. Thus perpendicular component of image velocity must be negative of that of the object, while component parallel to the mirror must be same (see adjacent figure).



→ In case the mirror is also moving, the image velocity problems should be first solved in the mirror frame and then converted to ground frame.

EXAMPLE: Find the velocity vector of the image when the ball is just about to hit the mirror (inclined at 37°). The ball was released from rest.



Solution: Speed just before hitting $= \sqrt{2gh} = \sqrt{2\times10\times5} = 10\,m/s$

Now taking components, $\vec{v} = 10\cos53°\,\hat{i} - 10\cos37°\,\hat{j}$

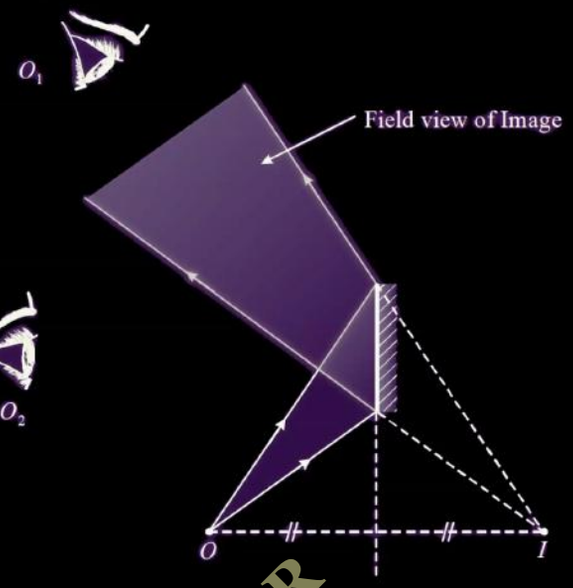
$\Rightarrow$ Image velocity $\vec{v}_i = 10\cos53°\,\hat{i} + 10\cos37°\,\hat{j}$

i.e $\vec{v}_i = 6\hat{i} + 8\hat{j}$ Ans.

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Field Of View Of An Image:

When you stand in front of a dressing table at some position, you are able to see image of several objects however not the image of every object that is being formed in the mirror. However if you shift your position, you are able to see images of some other objects. (For instance consider the adjacent figure, Here image of O is formed in the mirror however observer O_1 recieving the reflected light but O_2 is not recieving the reflected light. The region where an observer can be present so as to be able to see the image of an object is said to be the field of view of the image of the object.

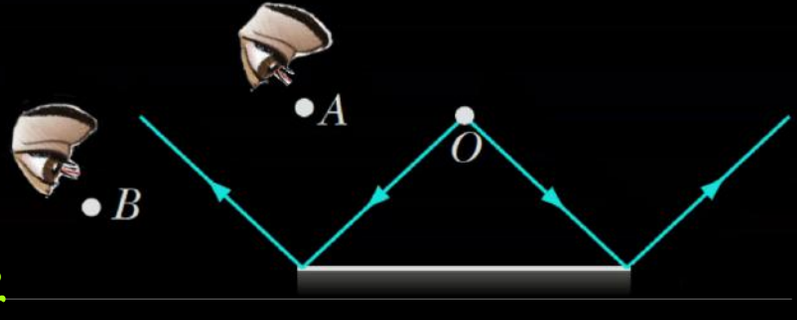


→ For finding field of view of an image in a plane mirror, we trace the rays from the object to the extremities of the mirror and find out the region of space enclosed by the reflected rays from the mirror. This is the field of view of the image of that point object.

→ For human observers to be able to see image of any object, divergent rays should fall on the eye lens. If convergent rays are incident on the eye after reflection, according to some authors, that point of observation should not be considered to be in field of view. (Though most authors don't mention this). (Actually for proper viewing of an image by human eye, it is not even sufficient to recieve divergent rays, rather point of divergence should be atleast 25 cm away, which is also known as least distance of distinct vision.

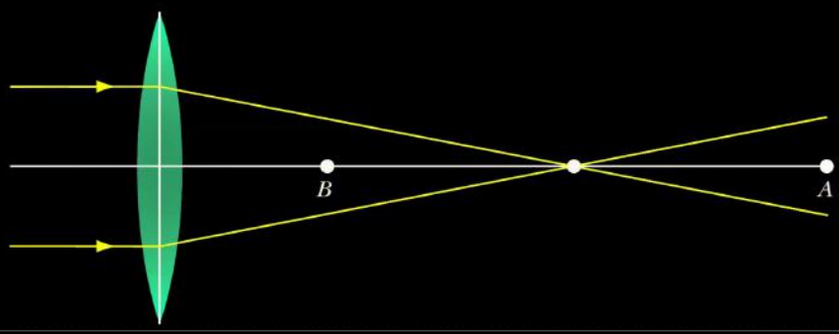
→ For a rectangular mirror, field of view will look like frustum of a pyramid and for a circular mirror it will look like frustum of a cone.

In the figure, A is in field of view but not B.



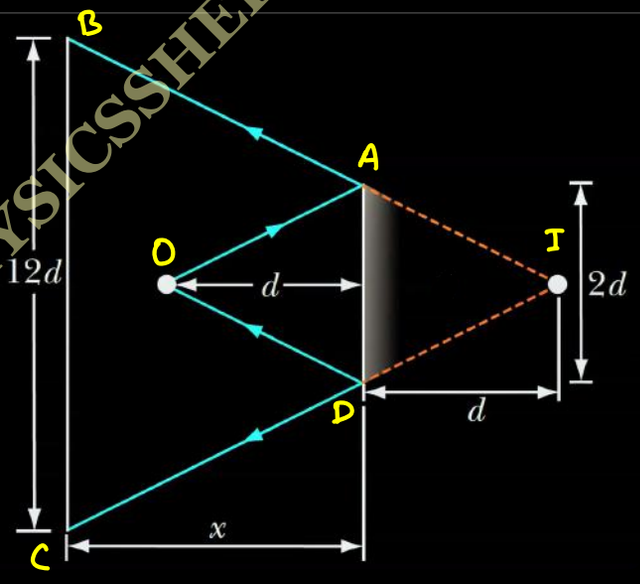
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→ For object at infinity, the image formed by the lens in figure will be visible to a human observer at A (Because he is in divergent pencil of rays) while not properly visible to observer at B, although, B will recieve rays which will cause an illumination on retina, however since B is in convergent portion of the pencil of rays, he can't form a proper image on retina (Retina is about 2.5 cm behind the eye lens).



Example:

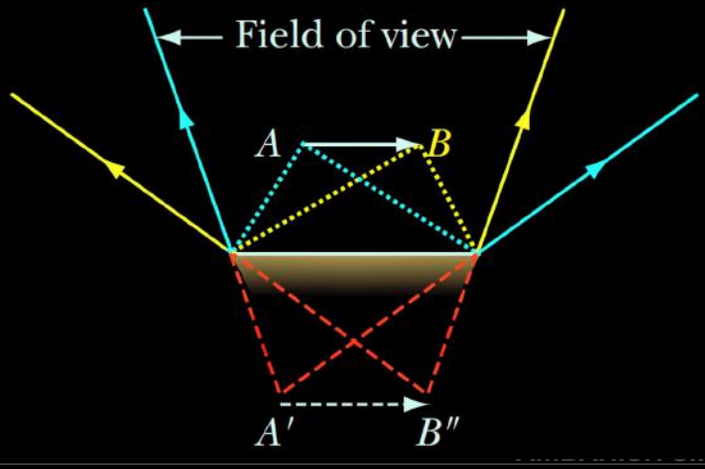
A mirror of length $2d$ is kept in front of a road as in Figure. A point object is kept at a distance d from the mirror. An observer is able to see the image till distance $12d$ while moving along the road. What is the distance x between the road and the mirror?



Solution: Considering the ray diagram $\Delta IAD \sim \Delta IBC$, thus, ratio of their altitudes from I must be in same ratio as the ratio of their sides, thus, $\frac{d}{d+x} = \frac{AD}{BC} = \frac{2d}{12d} \Rightarrow 6d = d + x \Rightarrow x = 5d$ Ans.

Field Of View Of An Extended Object In A Plane Mirror:

Intersection region of field of views of all points of an object in a mirror is called field of view of the image of the object. For locating the field of view of the object, we just draw the fields of view of its extreme particles and take their intersection. For instance in the shown

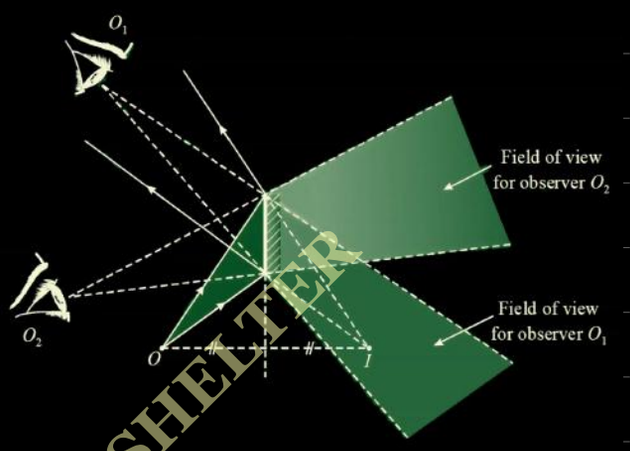


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figure, Turquoise rays define field of view of point A while yellow lines define field of view of B, In the region common to their fields of view we will be able to see the image of entire object.

Field Of View Of A Mirror For An Observer:

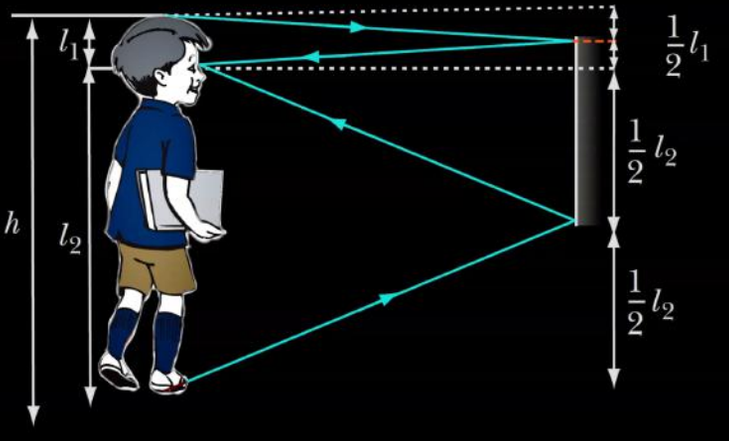
Region of virtual space (due to reflection in a mirror) that is visible from a point of view of an observer is the field of view of the observer in the image space. If any image is formed in in the field of view, it will be visible to the observers. For instance in the shown figure, image of O is in field of view of observer O_1 but not in field of view of observer O_2.



→ If the image of the eye in the mirror is visible from some point of space then by principle of reversibility that point will also be visible to the eye, thus to obtain the field of view mapping in object space, we can treat eye as the object and find field of view for that object.

Example: Find the minimum height of a mirror required to see one's complete Body in mirror.

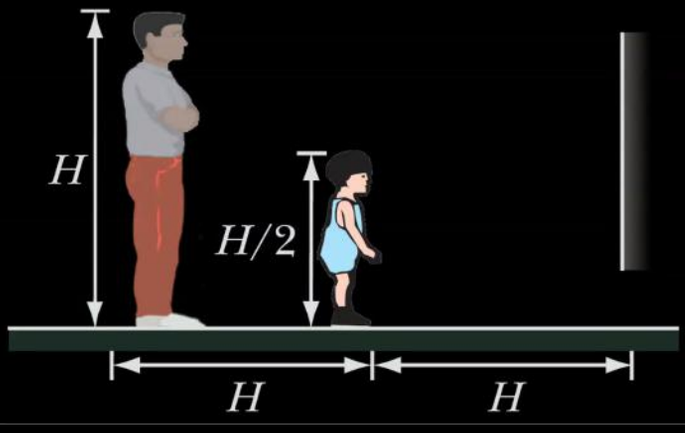
Solution: For head and toe to be visible, they should be in observer's field of view in the object space. So we take rays from the eyes to reflect in mirror and reach head and toe respectively (see figure) Now by similar triangles the portion of mirror utilized is from mid point of topmost point to the eyes to mid point of feet and eyes. Thus by geometry required height of mirror is $h/2$.



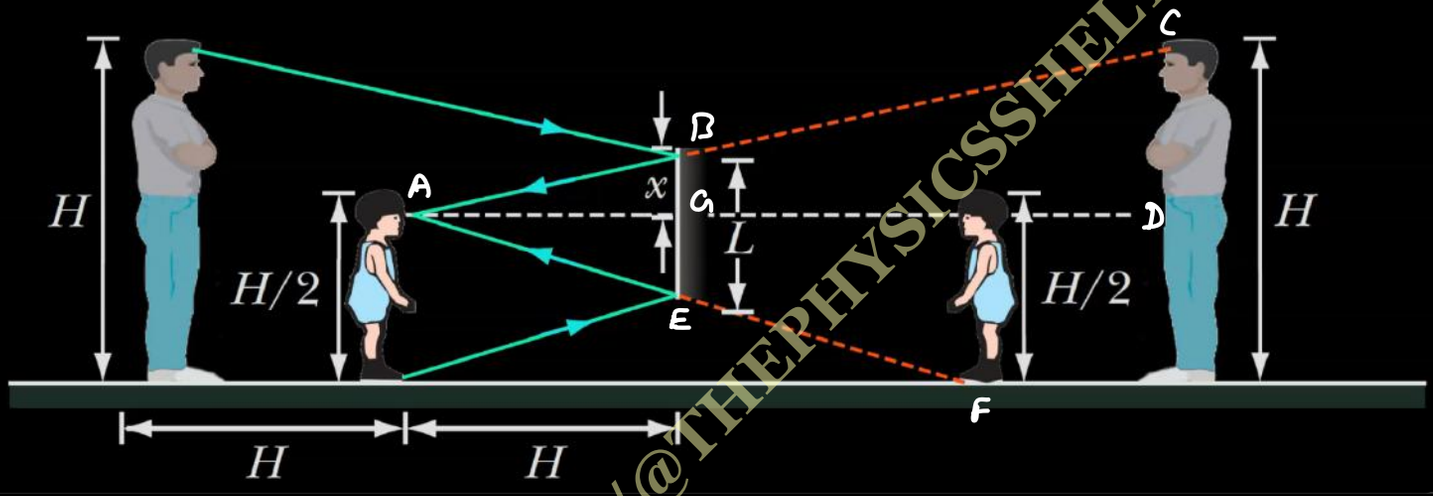
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Example:

A child and his father are standing in front of a mirror, find the required height of the mirror so that the child can completely see himself and top of his father's head in the mirror. Assume that eyes are at topmost point of the body.



Solution:



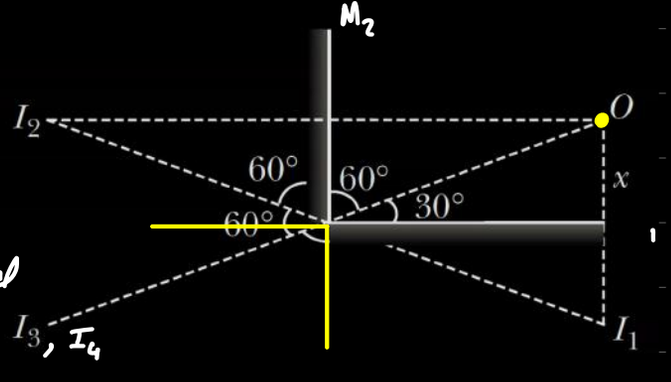
The mirror can be thought as a window in to image world. For anything to be visible it must be in the field of view of observer. (See figure above). The vertical coordinate of lower end of mirror should be half the height of the boy, i.e H/4 from ground. Also $\Delta AGB \sim \Delta ADC$ thus,

$\frac{x}{H/2} = \frac{H/3}{H} = \frac{1}{3} \Rightarrow x = H/6$, thus upper coordinate of mirror:

$\frac{H}{2} + \frac{H}{6} = \frac{2H}{3}$, thus required height of mirror: $\frac{2H}{3} - \frac{H}{4} = \frac{5}{12}H$ Ans.

Two Mirrors At Inclination: Multiple Image Formation:

Whenever an object is placed between two mirrors, the image formed by one mirror serves as an object for the other mirror provide it is formed in front of the other mirror. Thus several images are formed until image formed by one of the



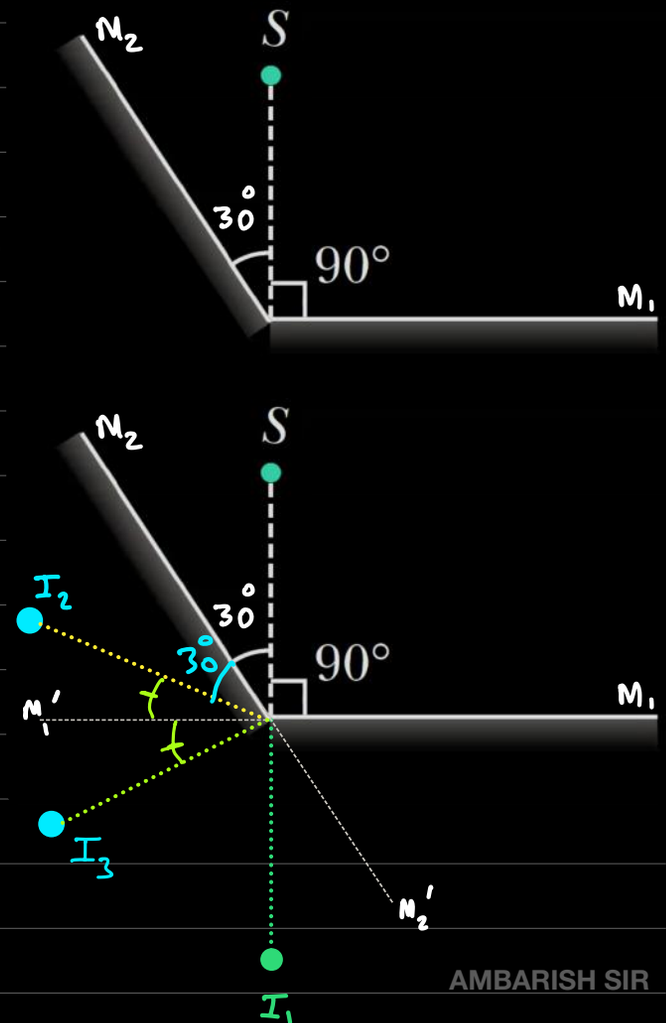
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mirrors is behind the other mirror (now no more reflection is possible). All such images lie on a circle with radius equal to the distance between the common edge of mirrors and the point object. For instance in the figure shown above, O is the object and I_1 and I_2 are its images due to direct reflection in horizontal and vertical mirrors respectively, Now since I_1 is in front of M_2, it is further reflected in M_2 to form I_4 similarly since I_2 is in front of M_1 it is further reflected to form I_3 (which happens to coincide with I_4). Now I_3 and I_4 are behind both the mirrors so they do not undergo any further reflection. Thus here total number of images appearing is 3 ($\because I_3$ & I_4 are coinciding) However you can check that if instead of 90° if the angle between the mirrors were 89° or 91°, you would be able to see 4 distinct images. The procedure described here can be used for finding image formation for any arbitrary case, however later on we will also be stating an algorithm for quickly finding number of images.

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EXAMPLE:

Compute the number of images for the shown two mirror system.



Solution: First of all direct images I_1 and I_2 are formed due to direct reflections from the mirrors. Now I_2 is formed in front of M_2 while I_1 is formed behind M_1 thus I_2 is further reflected from M_1 to form I_3 while I_1 is not reflected any further from M_2. Now I_3 is in the "shadow" region i.e behind both mirrors and can't be reflected any further. Thus only three images are formed by the system.

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From our experience of the previous examples we can state the following:

1. If the object is at angle θ in front of the mirrors, the image will also be at an angle θ behind mirrors.
2. All these images lie on a circle whose center is the point of intersection of the mirrors. The radius of the circle is equal to the distance between the point of intersection of the mirrors and the object.
3. The reflection will continue to take place till the image lies in front of the plane of the mirror. In other words, if reflected rays do not reach the next mirror again, further images will not form.

General rule for finding the number of image when the mirrors are inclined at an angle θ.

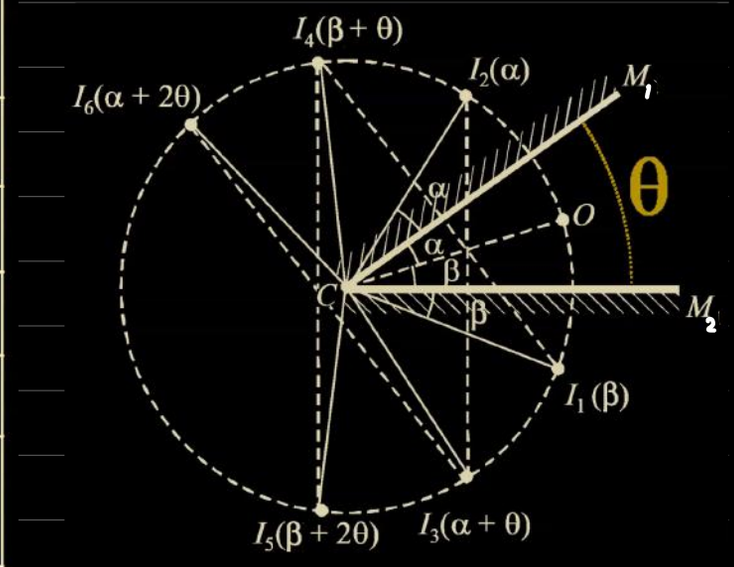
Let $m = 360/\theta$. Then

1. If m is an even integer, the number of images will be $(m - 1)$ irrespective of the position of the object.
2. If m is an odd integer, we have the following cases:
   Case-I: If the object is on angle bisector of the mirror, the number of images will be $(m - 1)$.
   Case-II: If the object is at any other place, the number of images will be m. We cannot apply this rule when $360/\theta$ is not an integer.

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Algorithm For Evaluating Number Of Images:

Image formed by mirror M_1	Image formed by mirror M_2
α	β
$\beta + \theta$	$\alpha + \theta$
$\alpha + 2\theta$	$\beta + 2\theta$
$\beta + 3\theta$	$\alpha + 3\theta$
upto the limit till this angle is less than 180°	upto the limit till this angle is less than 180°



→ If θ is angle between two mirrors M_1 and M_2 and an object O is positioned forming angles α and β with mirrors as shown in the figure. The figure shows the method for plotting successive images which leads to the Algorithm for getting image angular coordinates relative to respective mirrors. This algorithm is illustrated

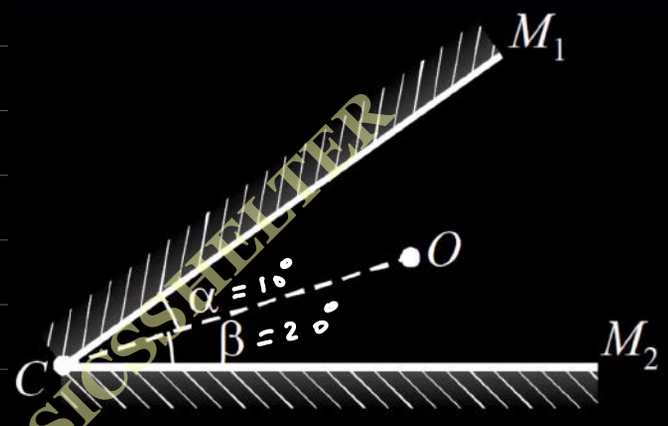
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in the table above.

→ For any object in front of the mirror the mirror will always form an image behind itself, thus angular coordinate of image can't be more than 180°.

→ If 0 + Sum of entries in the last row add up to 360° the images represented by last row are coincident and must be counted as a single image.

EXAMPLE: Using the Algorithm described in previous section calculate the number of images for $\alpha = 10°$ and $\beta = 20°$.



Solution: $\alpha + \beta = 30°$

S. No.	M_1		M_2
1	10°	+30 +30	20°
2	50°		40°
3	70°		80°
4	110°		100°
5	130°		140°
6	170°		160°
	STOP		STOP

Last row Calculation:

$170° + 160° + 30° = 360°$ thus images in last row coincide. Thus total number of images: $6 \times 2 - 1 = 11$

Ans.

SPHERICAL MIRRORS:

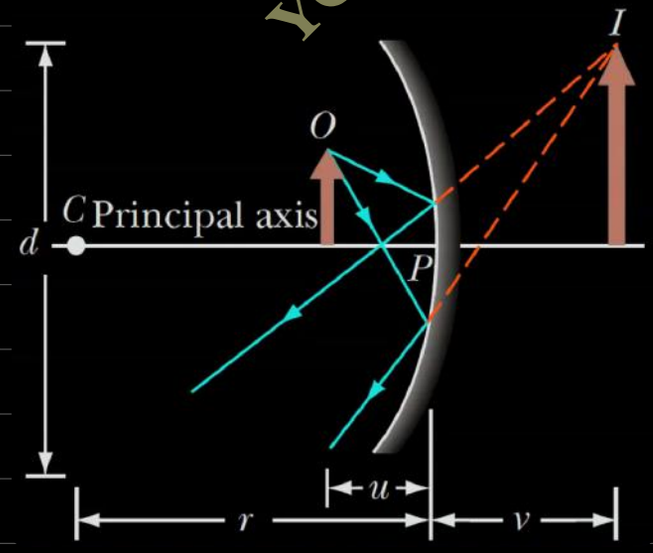
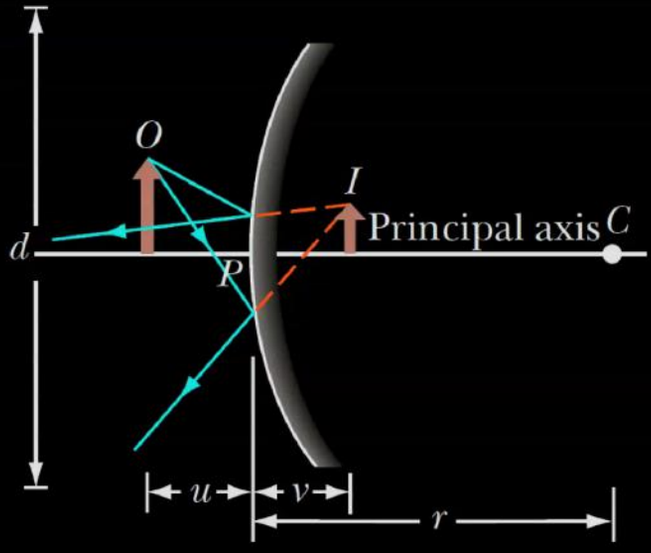


IMAGE FORMATION: CONCAVE MIRROR



CONVEX MIRROR

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TERMINOLOGY: Refer to the figure of the two types of mirrors shown above. Here:

① Aperture: The diameter of "face" of the mirror is called aperture. This is shown by dimension d in the figures

② Centre of Curvature: The mirrors can be considered as portions cut from a a sphere, the centre of such a sphere is called centre of curvature, this is marked C in the figures

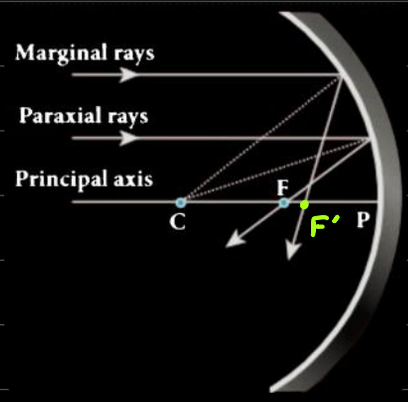
③ Pole: Mid point of the Mirror is called Pole (marked P in figures)

④ Principal axis: A line passing through the Pole and centre of curvature of a mirror is called principal axis.

⑤ Focus: A point where a ray parallel to principal axis cuts the principal axis after reflection from mirror is called focus (not shown in the figure).

→ Focus of a large aperture mirror is not well defined, this will be demonstrated in an upcoming section. For this reason we will mostly deal with small aperture mirrors (i.e $d << R$ where R is the radius of curvature of the mirror).

→ PARAXIAL RAYS: Rays which are close to the principal axis and make small angles of incidence on the mirror are called paraxial rays while the rays far from principal axis or making large angles of incidence are called marginal rays. As can be seen from the figure, focusing points of paraxial and marginal rays are different. (Marginal rays are focused nearer to the pole).



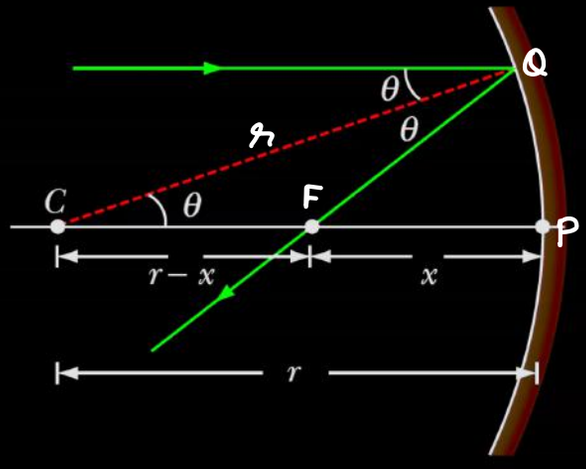
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FOCUSING POINT OF PARAXIAL AND MARGINAL RAYS || TO PRINCIPAL AXIS:

Let R be the radius of curvature of the mirror and x be the focal distance as shown in the figure. Now considering the triangle CFQ: $(R-x)/\sin\theta = R/\sin(180°-2\theta)$ (∵ Sine Law). Thus:

$$\frac{R-x}{\sin\theta} = \frac{R}{2\sin\theta\cos\theta}$$

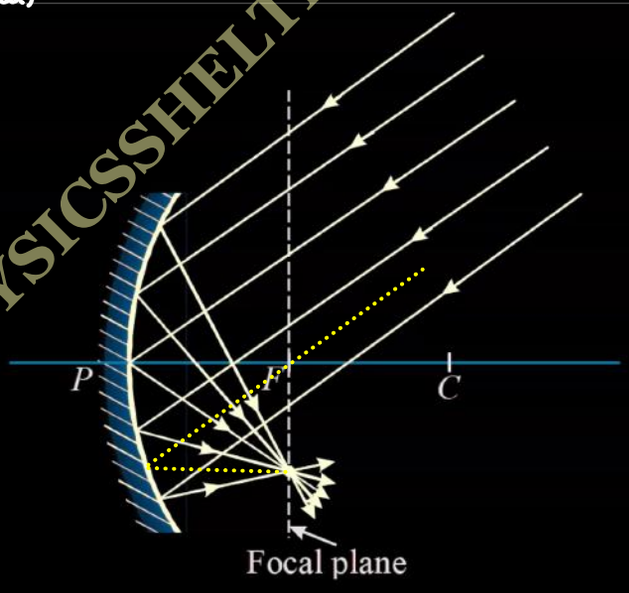
$$\Rightarrow 2(R-x)\cos\theta = R \Rightarrow \boxed{x = R - \frac{R}{2}\sec\theta}$$



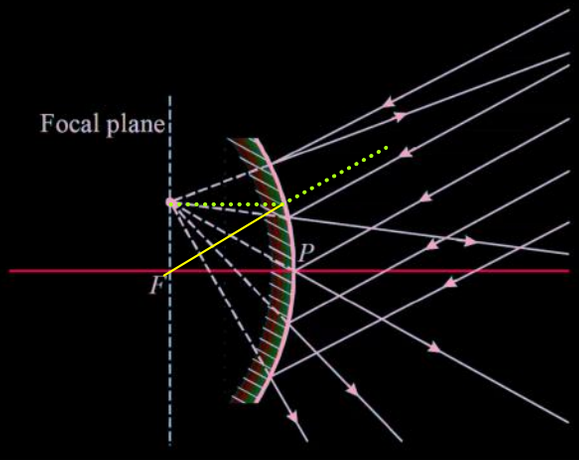
→ From this formula it is evident that focal distance depends on incidence angle. Now for paraxial rays θ is very small thus $\sec\theta \approx 1$ thus for paraxial rays, $\alpha = \pi/2$.

SPHERICAL ABERRATION: For formation of a sharp image of a point object, all reflected rays must get focused at the same point, however as seen in previous section, paraxial rays and marginal rays have different focusing points, this causes a blurring of image. This type of blurring is known as spherical aberration.

FOCAL PLANE: A plane passing through the focus of a mirror perpendicular to the principal axis is called focal plane. If a parallel beam of light (though not parallel to principal axis) strikes a mirror, it gets focused in the focal plane and the point of focusing is called secondary focus. To locate such a focal point we consider a ray passing through the focus and reflected it parallel to principle axis to intersect in the focal plane. This intersection point is taken as convergence point of all the rays in the beam.



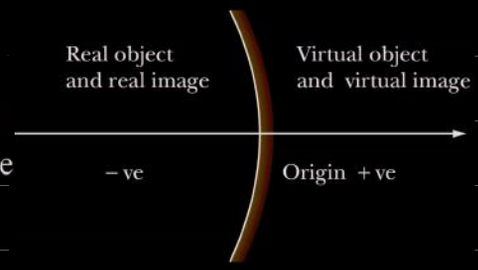
→ Similarly adjacent figure shows focal plane and focusing point for a convex mirror.



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New Cartesian Sign Convention:

1. The principal axis of the mirror is taken as x axis and the pole as origin.
2. The direction of the incident light is taken as positive direction of x axis.
3. The upward direction is taken as positive and downward direction as negative

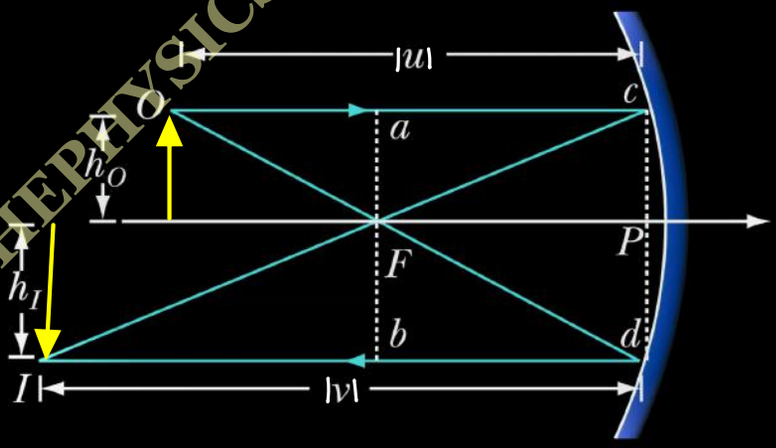


→ There also exist some other sign conventions like Gaussian Convention and Newtonian Conventions which are still used in many ophis textbooks, however for purpose of JEE we will always report results according to cartesian sign convention.

→ Image coordinate is typically represented by symbol v and object coordinate is typically represented by symbol u.

→ Focal length and radius of curvature are also signed values with sign decided according to incident light rule. (For concave mirror R & f are negative).

Mirror Formula: Consider adjacent ray diagram, here O is the object, the ray parallel to principal axis is reflected through focus and ray through focus is reflected parallel to principal axis to intersect at I i.e Image. h_I and h_o are assumed magnitudes of vertical coordinates of image and object. Now using similar triangles $OaF \sim dbF$ thus:



$$\frac{aF}{Fb} = \frac{-u+f}{-f} \quad (1)$$, Also, $acF \sim bIF$ thus:

$$\frac{aF}{Fb} = \frac{-f}{-v+f} \quad (2)$$ Now $(1)/(2) \Rightarrow \frac{-f}{-v+f} = \frac{-u+f}{-f}$

Now doing the Operation N/D+N : $\frac{-f}{-v} = \frac{-u+f}{-u} = 1 - \frac{f}{u}$ (3)

$$\Rightarrow f\left[\frac{1}{v} + \frac{1}{u}\right] = 1 \Rightarrow \boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}}$$

Magnification From Mirror: The ratio of image height to object height is called magnification, it is taken as negative if image is inverted. Now from equation (1)

$$m = \frac{-h_I}{h_o} = -\frac{Fb}{Fa} = \frac{f}{f-u}$$

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Now from equation ③, $\frac{f}{v} = \frac{f-u}{-u} \Rightarrow \frac{f}{f-u} = -\frac{v}{u}$ thus magnification from mirror: $\boxed{m = -v/u}$

NEWTON'S FORMULA: In the Newtonian convention, object and image coordinates (x and y respectively) are considered from focus instead of pole. Thus $x = u - f$ and $y = v - f$, now using $u = x + f$ and $v = y + f$ in mirror formula:

$$\frac{1}{x+f} + \frac{1}{y+f} = \frac{1}{f} \Rightarrow (x+f)(y+f) = (x+y+2f)f$$

$$\Rightarrow xy + (x+y)f + f^2 = (x+y)f + 2f^2 \Rightarrow \boxed{xy = f^2}$$

→ This formula means x and y must always have same sign ∵ f^2 is always positive, thus object and image always lie on same side of focus.

→ Magnification in Newtonian convention: $m = -v/u = -\left(\frac{f+y}{f+x}\right)$

Now putting $y = f^2/x$, $m = -\left(\frac{f + f^2/x}{f+x}\right) = \frac{-f(1+f/x)}{x(1+f/x)} = -f/x$

Thus $m = -f/x$ also $x = f^2/y \Rightarrow m = -y/f$ thus:

$$\boxed{m = -f/x = -y/f}$$

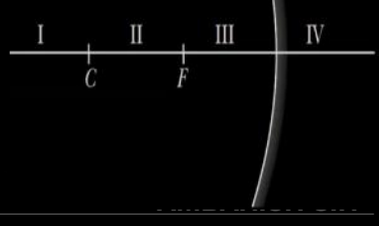
→ Newton's formula $xy = f^2$ looks much more elegant than the mirror formula and simplifies the calculations greatly in many problems. (Reduces the problems to mental calculations).

NATURE OF OBJECT AND IMAGE FOR CONCAVE MIRROR:

Concave Mirror: Nature of Object and Image

Object	Image	Nature	Magnification	Erect/Inverted
I	II	Real Object & Real Image	Diminished	Inverted
II	I	Real Object & Real Image	Enlarged	Inverted
III	IV	Real Object & Virtual Image	Enlarged	Erect
IV	III	Virtual Object & Real Image	Diminished	Erect

CONCAVE MIRROR.



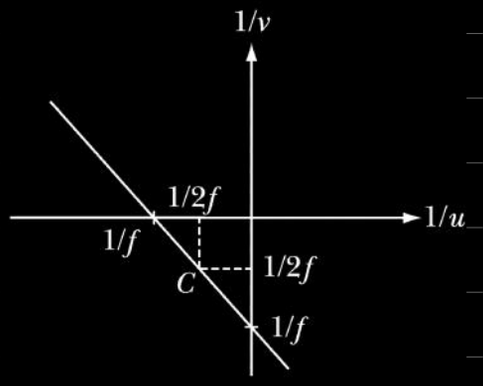
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→ $u<0$ ⇒ Real object $u>0$ ⇒ Virtual object

→ $v<0$ ⇒ Real image $v>0$ ⇒ Virtual Image

→ This table can be quick inferred from graph between $1/v$ and $1/u$

→ $\frac{1}{v}$ vs $\frac{1}{u}$ for a concave mirror is shown in adjacent figure. It is a straight line of negative slope and intercept-. Note that f is a negative number thus intercept $1/f$ represents a negative number.



Nature Of Object And Image For Convex Mirror:

Convex Mirror: Nature of Object and Image

Object	Image	Nature	Magnification	Erect/Inverted
I	II	Virtual Object & Virtual Image	Diminished	Inverted
II	I	Virtual Object & Virtual Image	Enlarged	Inverted
III	IV	Virtual Object & Real Image	Enlarged	Erect
IV	III	Real Object & Virtual Image	Diminished	Erect



Example :

A tarantula of height h sits cautiously before a spherical mirror whose focal length has absolute value | f | = 40 cm. The image of the tarantula produced by the mirror has the same orientation as the tarantula and has height h' = 0.20h.

(a) Is the image real or virtual, and is it on the same side of the mirror as the tarantula or the opposite side?

(b) Is the mirror concave or convex, and what is its focal length f, sign included? (c) How far is Tarantula from the mirror?

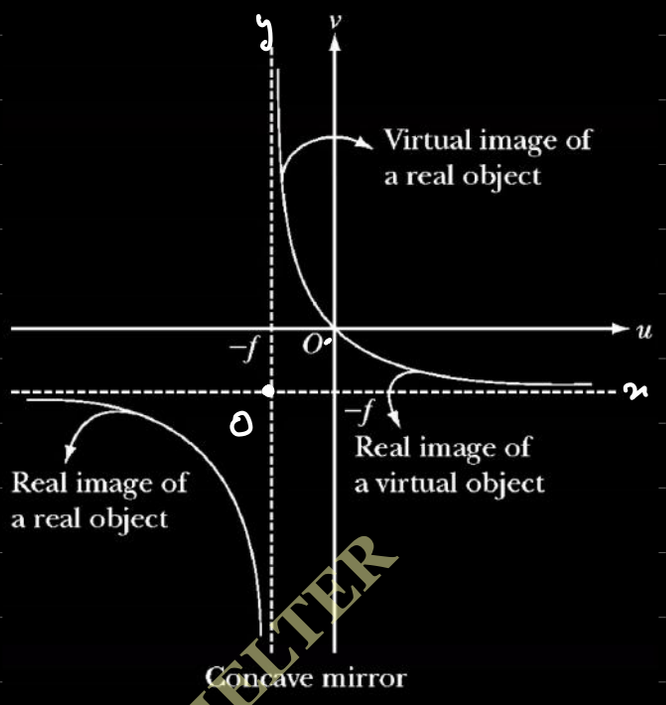
Solution: Tarantula is a real object and we are getting a diminished erect image of a real object thus our mirror must be a convex mirror. The image must be virtual i.e. on the opposite side of the mirror, thus $f = +40$ cm. Now:

$\frac{1}{v}+\frac{1}{u}=\frac{1}{40}$ and $\frac{-v}{u}=0.2$, solving we get: $u=-160$ $v=+32$

Thus spider is 160 cm in front of the mirror.

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u Vs v And x Vs y Graphs For Concave Mirror:



Recall the Newton's Formula $xy = f^2$ for spherical mirrors, thus $y = f^2/x$ which is the equation for a rectangular hyperbola. Also $u = f + x$ and $v = f + y$ thus u, v graph can be obtained by suitably shifting the x-y graph. Also note that f is a negative value, thus origin of u-v graph is shifted to $(-f, -f)$ wrt x-y graph. (When $x = -f$, $u = 0$ similarly when $y = -f$, $v = 0$).

→ Think, $xy = f^2 \Rightarrow y = \frac{f^2}{x} \Rightarrow v - f = \frac{f^2}{u-f} \Rightarrow v = \frac{f^2}{u-f} + f$

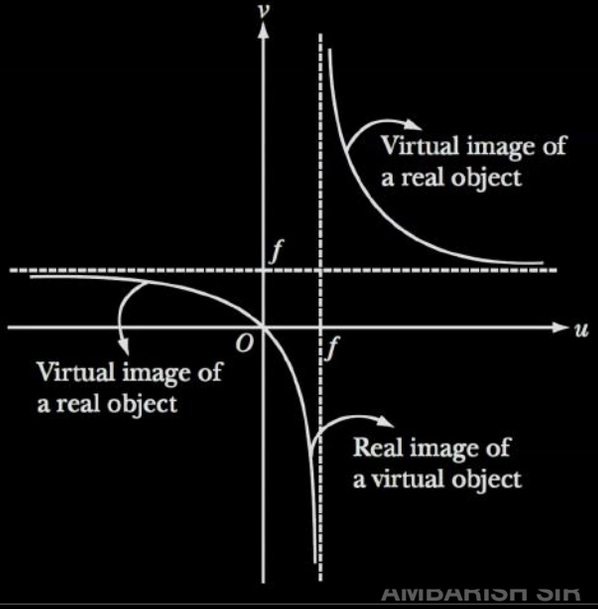
This is similar to $y = \frac{f^2}{x-f} + f$ thus u-v graph is shifted by f units rightward and upward relative to the origin but since f is negative we can say that it is shifted $-f$ units downward and leftward relative to origin, or if we do not wish to move the graph, we must move the origin rightward and upward by $-f$.

x-y Graphs And u-v Graphs For Convex Mirror:

For convex mirror also we have $xy = f^2$ However, f is positive for convex mirrors. Rearranging the equation we get:

$$y = \frac{f^2}{x} \Rightarrow v - f = \frac{f^2}{u-f}$$

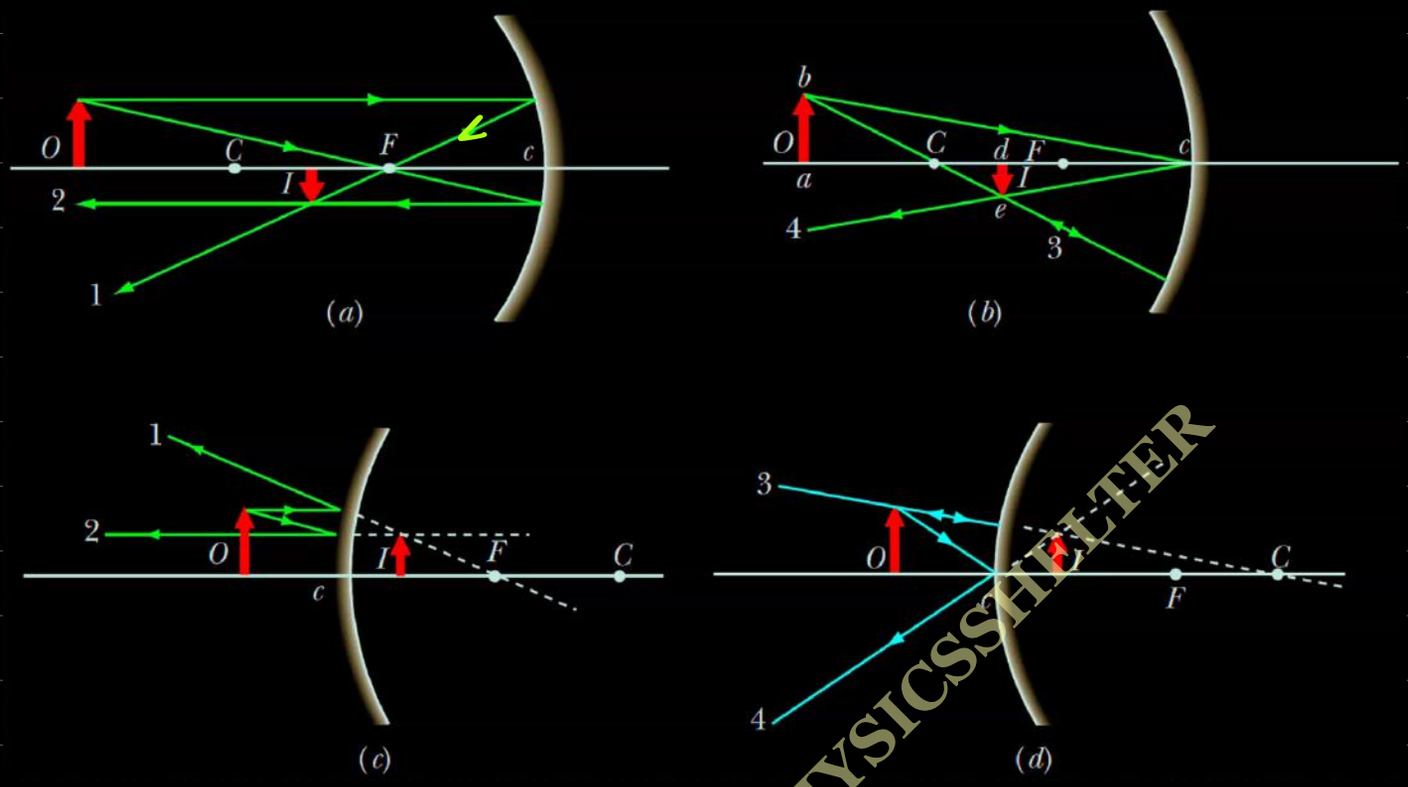
$\Rightarrow v = \frac{f^2}{u-f} + f$ thus uv graph is shifted by f units rightward and upward relative to origin or if graph is to be kept fixed, origin must be shifted f units



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downwards and leftwards. This is shown in the graph above.

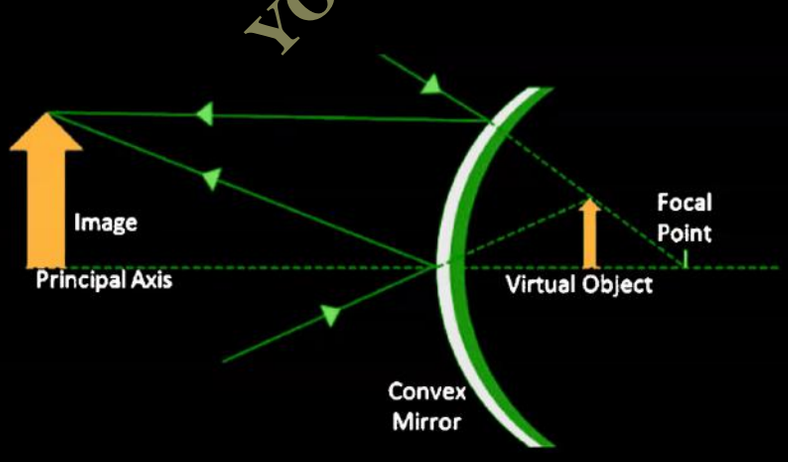
Ray Tracing For Spherical Mirrors:

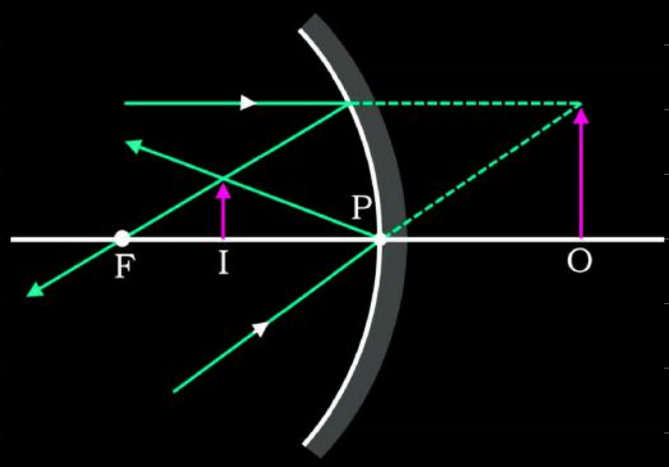


For finding image of an object we generally use any two of the following four rays:

- A ray passing through focus is reflected parallel to principal axis
- A ray parallel to principal axis is reflected through focus
- A ray through centre of curvature retraces its path upon reflection
- A ray incident on pole is reflected at same angle with principal axis.

Ray Tracing For Virtual Objects:





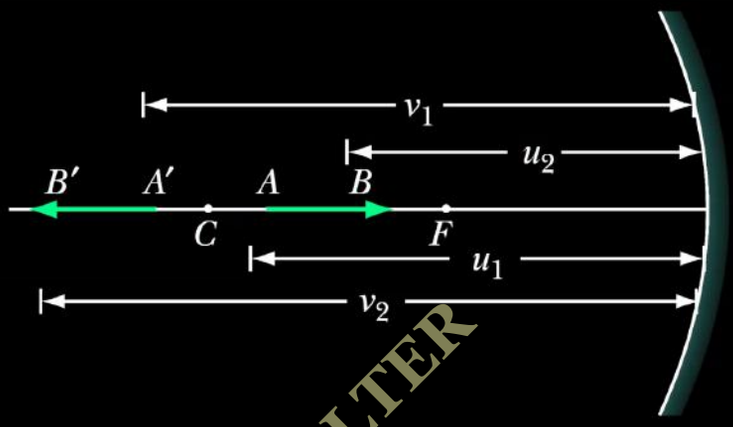
- Recall that a virtual object is formed when a convergent pencil is obstructed by a mirror or a lens.

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→ The procedure for ray tracing is similar to that for real objects. Two convenient rays from convergent pencil are chosen and reflected according to rules mentioned in previous section.

Longitudinal Magnification:

The ratio of image span to the object span for an object lying along the principal axis is called longitudinal magnification. Since such an image is always longitudinally inverted, longitudinal magnification is always negative.



Consider the shown image, here AB is the object with image being A'B'. Now longitudinal magnification is: $m_L = -A'B'/AB$, thus:

$$m_L = -\left(-v_2 - (-v_1)\right) / \left[-u_1 - (-u_2)\right] = (v_2 - v_1 / u_2 - u_1)$$

To find magnification for relatively longer objects (dimensions comparable to focal length), we must locate the v_2 and v_1 separately using the mirror formula.

→ For very small objects, we may write $m_L = dv/du$ or in Newtonian notation, $m_L = dy/dx$. Now $xy = f^2$

$$\Rightarrow \ln x + \ln y = \ln f^2 \Rightarrow \frac{1}{x} + \frac{1}{y}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -y/x = \frac{-f^2}{x^2}$$

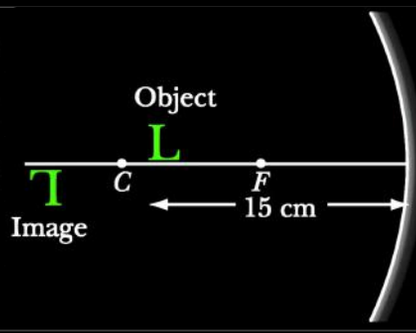
Now Recall that transverse magnification was $-f/x = -y/f$, thus we can see that $m_L = -m^2$. Thus:

$$\boxed{m_L = -m^2 = -\frac{v^2}{u^2} = -\frac{y}{x} = -\frac{f^2}{x^2} = -\frac{y^2}{f^2}}$$

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Example :

An L-shaped object with dimensions 1 cm each is kept on the principal axis of a concave mirror of focal length 10 cm as shown in the figure. Its edge is 15 cm away from the pole. Find the location, shape, and size of the image.



Solution: $x = -5cm$, $y = f^2/x = 100/-5 = -20cm$. Thus image is formed at a distance of 20 cm from focus on 30 cm from pole (Assuming object to be small sized one). Now longitudinal magnification, $m_L = -y/x = -(-20)/(-5) = -4$ and $m = -f/x = -y/f = -2$ Thus image is formed within a rectangle of height 2cm and width 4cm and is both laterally and longitudinally inverted.

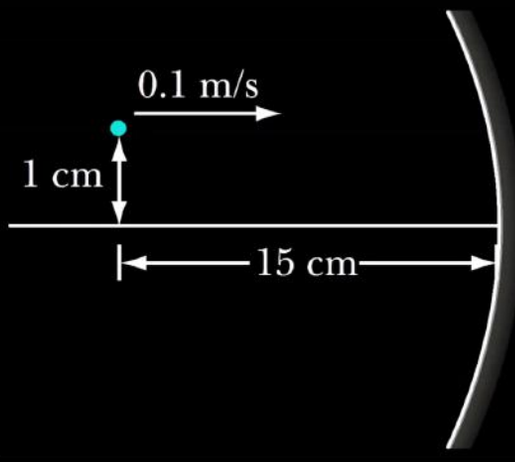
→ Exact method: $x_1 = -5$, $x_2 = -4$ $\Rightarrow$ $y_1 = f^2/x_2 = -20$, $y_2 = f^2/x_2$ $\Rightarrow$ $y_2 = -100/4 = 25cm$, Now $m_L = y_2 - y_1/(x_2 - x_1) = -5$, thus width of image is 5cm. Since 1cm is not very less compared to focal length and the distance from focus (5cm) is also comparable to 1cm, we have a significant errors in our approximation of assuming a small sized object.

Velocity Of The Image:

Longitudinal Velocity: Recall that $xy = f^2$

$\Rightarrow \ln x + \ln y = \ln f^2 \Rightarrow \frac{1}{x}\frac{dx}{dt} + \frac{1}{y}\frac{dy}{dt} = 0$

$\Rightarrow \frac{dy}{dt} = -\frac{y}{x}\frac{dx}{dt} = m_L\left(\frac{dx}{dt}\right)$, thus component of image velocity parallel to principal axis is simply longitudinal magnification times the object velocity along the principal axis.



Now let us find velocity perpendicular to the principal axis. Let h_i and h_o represent transverse coordinates of image and object. Then:

$h_i = m h_o = -\frac{f}{x} h_o$

$\Rightarrow d h_i/dt = -f\left[\frac{1}{x}\cdot\frac{dh_o}{dt} + h_o\left(-\frac{1}{x^2}\,dx/dt\right)\right]$

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Expanding, $\dot{h_i} = -f/x\ \dot{h_o} + \frac{h_o}{f}\frac{f^2}{x^2}\dot{x}$

Now if V_L and V_T represent longitudinal and transverse velocities of image and U_L and U_T represent longitudinal and transverse velocities of object then:

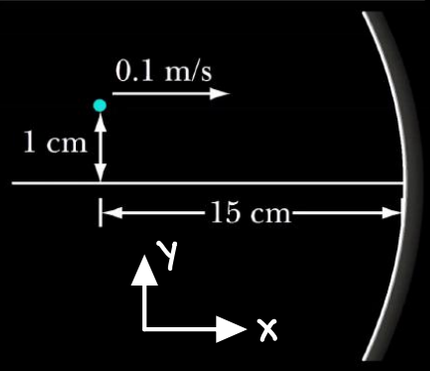
$$\boxed{V_L = m_L U_L}$$

$$\boxed{V_T = m U_T + \frac{h_o}{f} m^2 U_L}$$

→ Remember the sign convention: U_L is positive if moving along incident light and negative if against incident light.

EXAMPLE:

An object is traveling toward a concave mirror of focal length 10 cm. It moves with a velocity of 0.1 m/s parallel to the principal axis (See Figure). When it is at a distance of 15 cm from the mirror at a height of 1 cm from the principal axis, find the velocity of image in x and y direction?



Solution: $x = -5 \Rightarrow y = 100/-5 = -20\ cm$, $m_L = -y/x = -4$, $m = -f/x$

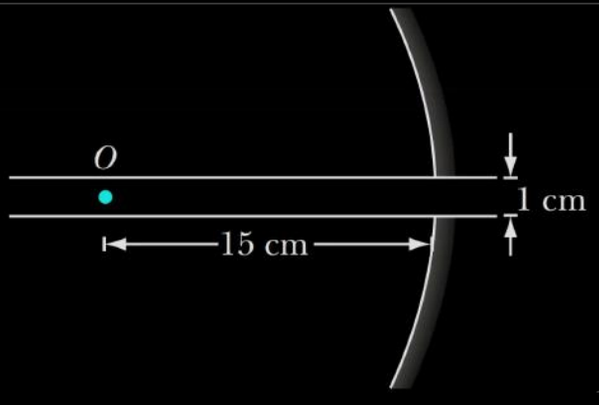
$\Rightarrow m = -2 \quad (\because f = -10)$ Now: $V_L = m_L U_L = -4 \times (+0.1) = -0.4\ ms^{-1}$

Also: $V_T = m U_T + \frac{h_o}{f} m^2 U_L = -2 \times 0 + \frac{1}{-10} \times 4 \times 0.1 = -0.04\ ms^{-1}$

Thus image velocity $\vec{V} = -0.4\hat{i} - 0.04\hat{j}$ Ans.

SPLITTING A MIRROR:

An image is formed by a mirror due to reflection of rays from its entire surface and recall that we could locate the image using any two convenient rays reflected from the mirror. Now if a mirror were to be split, still each piece of the mirror can receive several rays and form an image. For the purpose of location and size of the image,

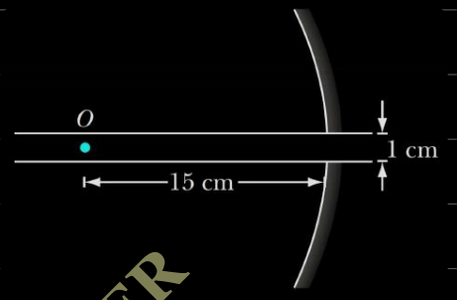


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each piece of the mirror behaves just like the full mirror. However since piece of a mirror has less light collecting area, it forms the image with lesser number of rays and hence the image is dimmer (or less intense).

Example:

A mirror of focal length 10cm is broken into two parts and these parts are separated by a distance of 1 cm as shown in the figure. Find the location of the images formed by the two parts of mirror. The object is midway between the two principal axes.



Solution: Each part of mirror behaves like a full mirror for purpose of location and size of image. We will locate the image due to the upper mirror and the image due to lower mirror can be found by symmetry.

Now for upper mirror, $x = -5\,cm$, $h_o = -0.5\,cm$, thus $y = f^2/x$

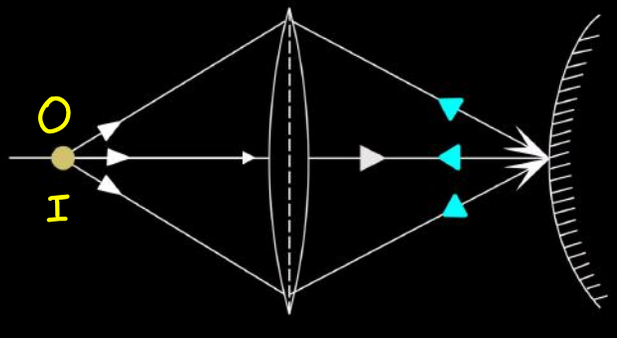
$\Rightarrow y = -20\,cm$, $m = -f/x = -2$ thus $h_i = -2 \times -0.5 = 1\,cm$.

Thus for the upper mirror, image is located at 30 cm from pole and 1 cm above the principal axis and for the lower mirror, by symmetry, the image is located 1 cm below the principal axis and 30 cm from pole. Ans.

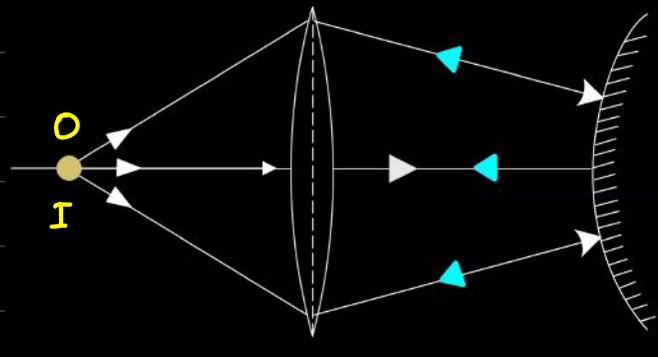
→ Think: What is the separation between the two images?

Ans: 3 cm (Remember to account for the fact that the two principal axes are separated by 1 cm).

Auto Collimation Of A Point On Principal Axis Of Mirror System:



Ray Exchange



Ray Reversal

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If the image of an object is formed at the location of the object itself, we call the condition as autocollimation. For a mirror system such as the one shown in the system, autocollimation can be achieved for a point object on the axis in following two ways:

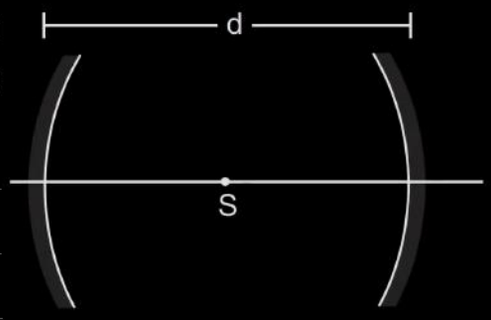
1. Ray Reversal: If the rays fall normally on the mirror, they begin to retrace their path and hence by reversibility principle, they will again intersect at the object. (Rays falling normally is equivalent to saying that rays are forming a virtual object at centre of curvature.
2. Ray Exchange: Here a ray below the principal axis returns along the ray above the principal axis and vice-versa. So effectively sort of a reversal.

→ Here we are talking about autocollimation using a single reflection. There exist other geometries for auto-collimation involving multiple reflections also.

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EXAMPLE:

A point source S is placed midway between two converging mirrors having equal focal length f as shown in figure. Find the values of d for which only one image is formed. (Ignore any image formed at infinity).



Solution: We can of course solve this problem using algebra, however if we think carefully we can solve this problem very quickly. By symmetry of the geometry we can say that if only one image has to be formed, it must be formed at S itself. So one obvious answer would be to have S as the common centre of curvature thus $d = 4|f|$ cm (This is one possibility). Now we do not want any other image to form not counting the one at infinity so let us think what happens if the image is formed at infinity. In that case S must be at the focus of the mirror. The rays reflected from one mirror will become parallel and again reflect from the other mirror to once again get focused at S itself, thus other possibility

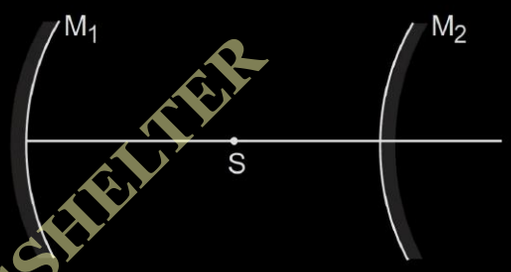
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is to have S as the common focus. In that case, $d = 2|f|$. (Another possibility would be having $d=0$, but that's physically irrelevant). One can easily see here that by choosing S to be anything else, there will definitely be other intermediate images.

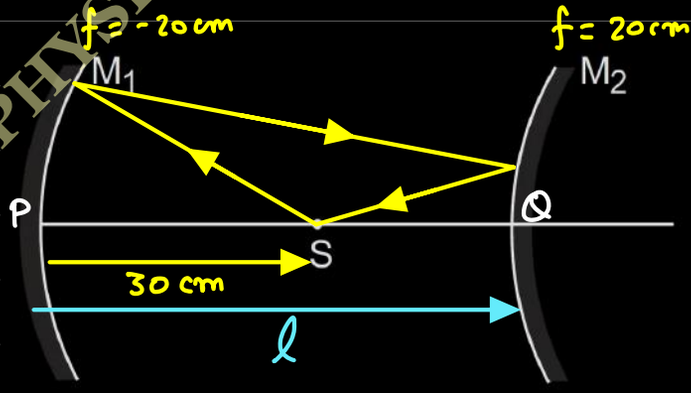
→ If the mirrors had different focal lengths we could achieve a similar condition using $d = |f_1| + |f_2|$ or $|2f_1| + |2f_2|$

EXAMPLE :

A converging mirror M_1, a point source S and a diverging mirror M_2 are arranged as shown in figure. The source is placed at a distance of 30 cm from M_1. The focal length of each of the mirrors is 20 cm. Consider only the images formed by a maximum of two reflections. It is found that one image is formed on the source itself. (a) Find the distance between the two mirrors. (b) Find the location of the image formed by the single reflection from M_2.



Solution: The ray diagram considering first reflection from M_1 and second reflection from M_2 is shown in the figure. (If we consider first reflection from M_2 and then from M_1 still the ray diagram is almost same except that all arrow directions reverse).



Now for M_1 : $xy = f^2$

$\Rightarrow -10xy = 20^2 \Rightarrow y = -40$ i.e 60 cm to the right of P.

Now for M_2, $u = 60 - \ell$, $v = -(\ell - 30)$, thus:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{30-\ell} + \frac{1}{60-\ell} = \frac{1}{20} \Rightarrow \ell = 50 \text{ or } 0$$

(Zero is irrelevant answer), thus separation between mirrors is 50 cm Ans.

Now for direct image from M_2 :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad , \quad u = -20 \quad , \quad \frac{1}{v} + \frac{1}{-20} = \frac{1}{20} \Rightarrow v = +10$$

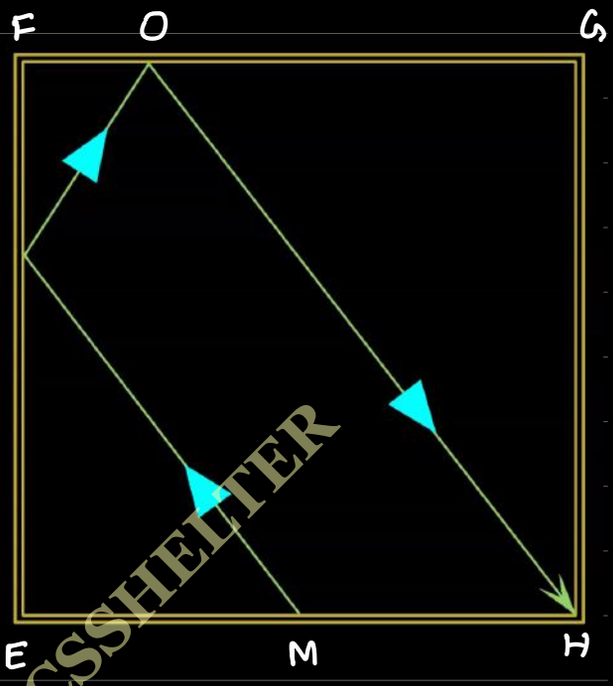
Thus image is formed 10 cm to the right of M_2 Ans.

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→ Think: What if 3 reflections were allowed for autocollimation?

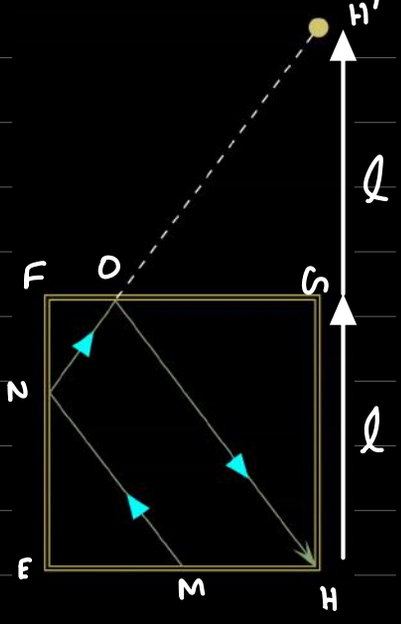
Ans: In that case, image from one mirror can be formed either at pole or centre of curvature of the other mirror.

Example: On a Carrom Board (shown schematically), A coin (assumed to be a point like object) starts from M and reaches H after making two elastic collisions as shown). M is the mid point of EH. If the collisions lead to mirror like reflections, find the angle of $\vec{MN}$ with $\vec{ME}$.



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Solution: Here we use the fact that if reflected ray is passing through a point then the incident ray must be passing through the mirror image of that point. Thus $\vec{NO}$ must be directed towards H' i.e image of H in mirror FG.

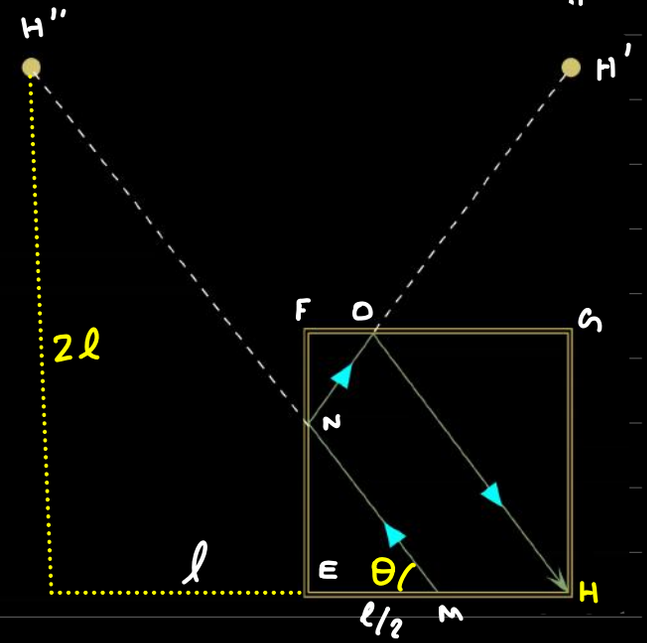


Similarly, if reflected ray passes through H', then incident ray must be directed along its mirror image in the mirror FE, i.e towards H''. Thus MN is directed along MH''.

Now using simple trigonometry:

$$\tan\theta = \frac{2l}{l + l/2} = 4/3$$

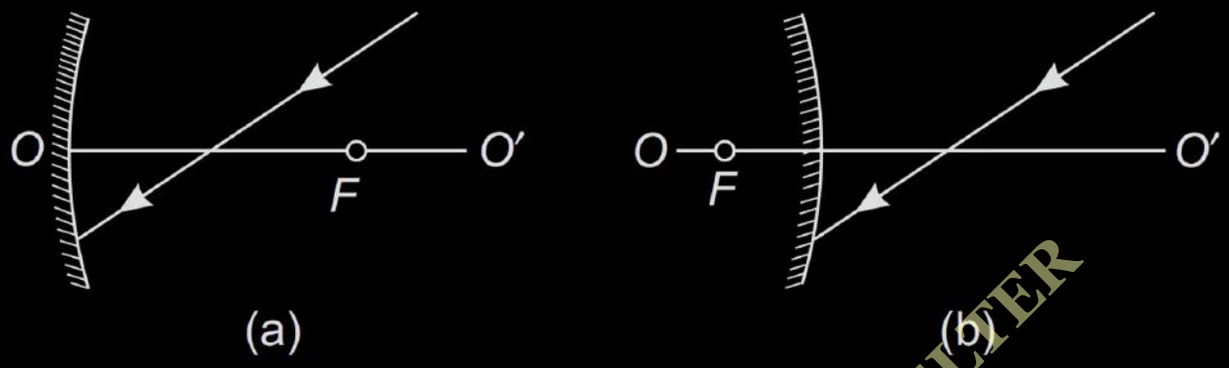
Thus the required direction of striking is along $\theta = 53^\circ$ approximately.



Think: How would you approach the problem if after three reflections you wanted to reach E.

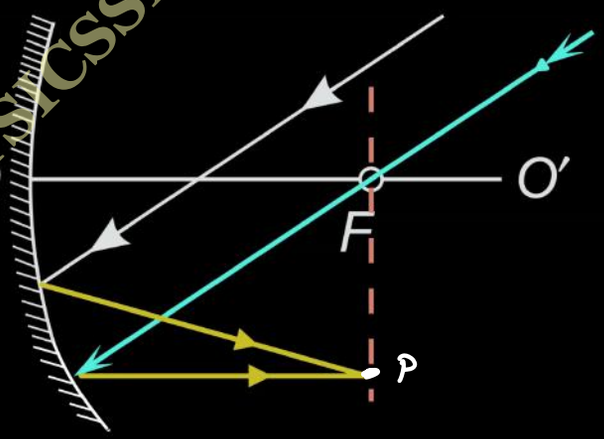
Example: Irodov: 5.26a:

By means of plotting find the path of reflected rays for the shown incident rays. (Paraxial assumption).

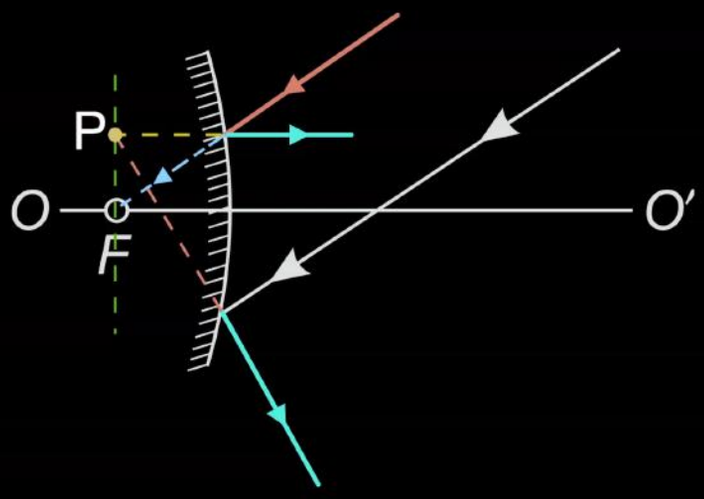


(a) (b)

Solution: (a) Imagine the ray to be part of a bundle of parallel rays. This bundle must focus in the focal plane. Accordingly we take a ray in the bundle passing through the focus reflected parallel to the principal axis to cut focal plane at P. The reflected ray thus then pass through P (see figure).

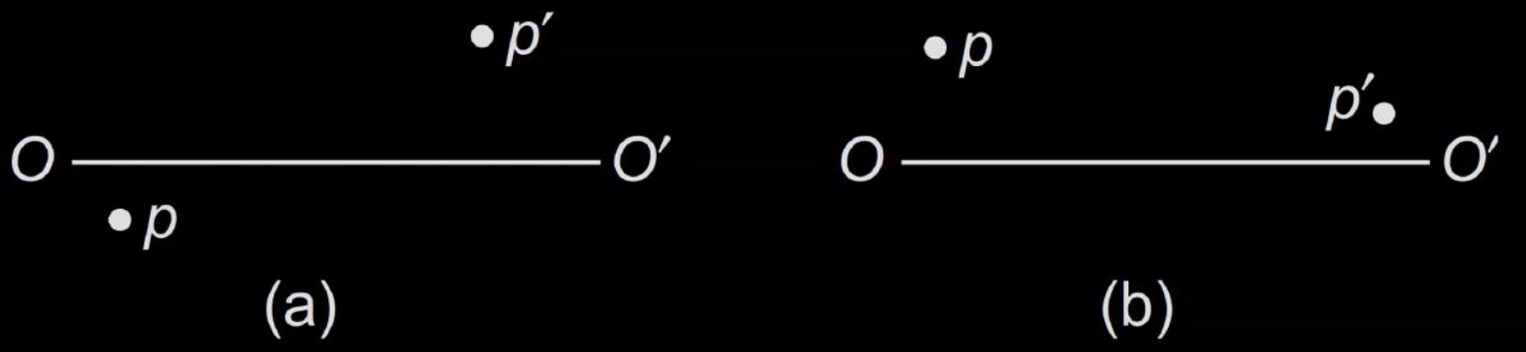


For convex mirror also we use a similar idea. Assume member of the bundle directed towards focus and reflect it parallel to principal axis and extend the reflected rate to intersect focal plane at point P, now we reflect the incident ray away from P to get our required ray.



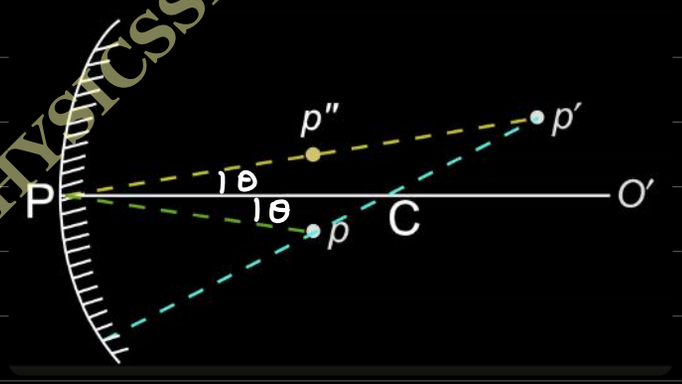
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Example: Irodov 5.26b:



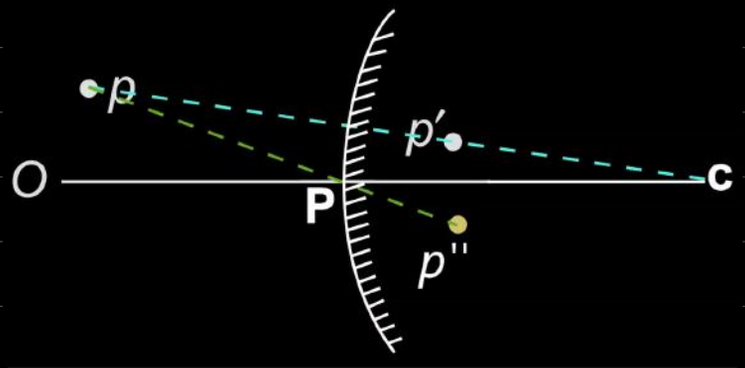
Locate the position of the mirror and its focal point for above cases. P and P′ are conjugate points (Object and image (interchangeable)).

Solution: (a) We use the fact that line joining the conjugate points passes through the centre of curvature to locate the centre of curvature and since the ray incident at pole is equally inclined to principal axis, this ray must pass through water image of p (p"). Thus by connecting p′ and p" we get the pole on the principal axis (see figure).



Similarly we obtain the diagram for second case.

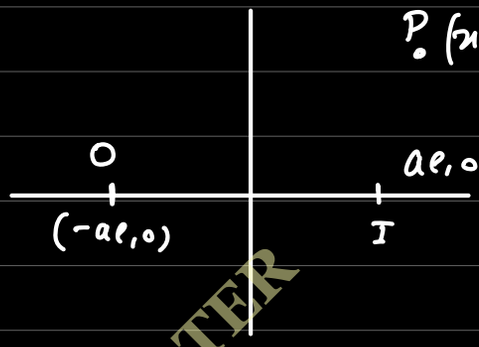
→ Note that we cannot comment on the nature of the mirror. Any face could be polished. To exactly distinguish the polished side we need to know whether object is real or virtual.



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Example: Using Fermat's principle of stationary path, find the equation of a mirror if object is located at $(-ae, 0)$ and image is located at $(+ae, 0)$. (According to stationary path principle all paths of light from object to image take equal times). (Both object and image are real).

Solution: Let (x, y) be a general point on the mirror where rays from object are reflected. Then, sum of distances from O and I of the incident and reflected line segments: $OP + PI$:



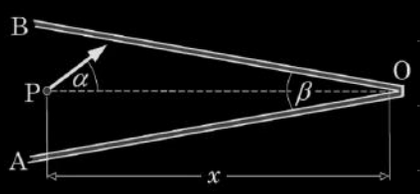
$$\sqrt{(x+ae)^2 + y^2} + \sqrt{(x-ae)^2 + y^2} = C = 2a \text{ (say)}.$$

This represents an ellipse with foci $(-ae, 0)$ and $(+ae, 0)$.

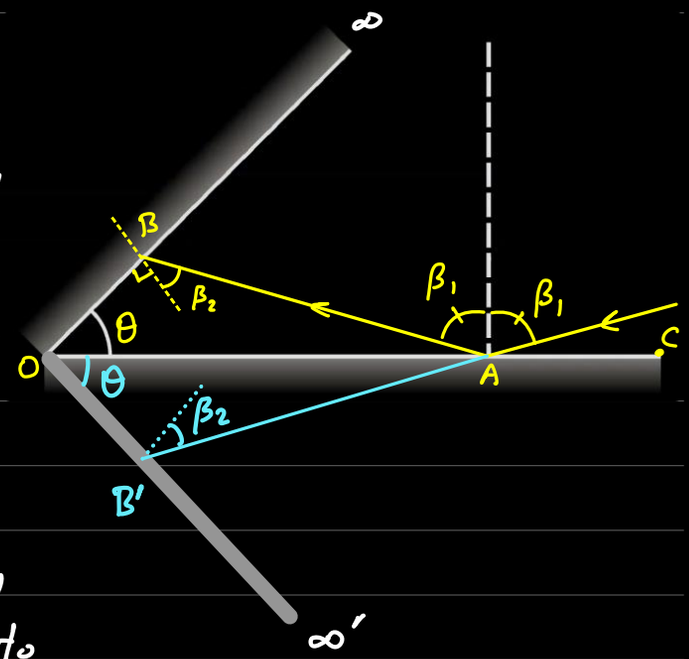
Example: Pathfinder: Build-6:

In free space, a particle is projected from a point P on axis of a fixed rigid cone AOB, at an angle $\alpha = 37^\circ$ with the axis (see figure). Distance of point P from the apex O is $x = 10$ cm and the apex angle of the cone is $\beta = 20^\circ$. All the collisions of the ball with the cone are perfectly elastic.

(a) Find the distance of closest approach of the ball from the apex.
(b) How many times will the ball collide with the cone?



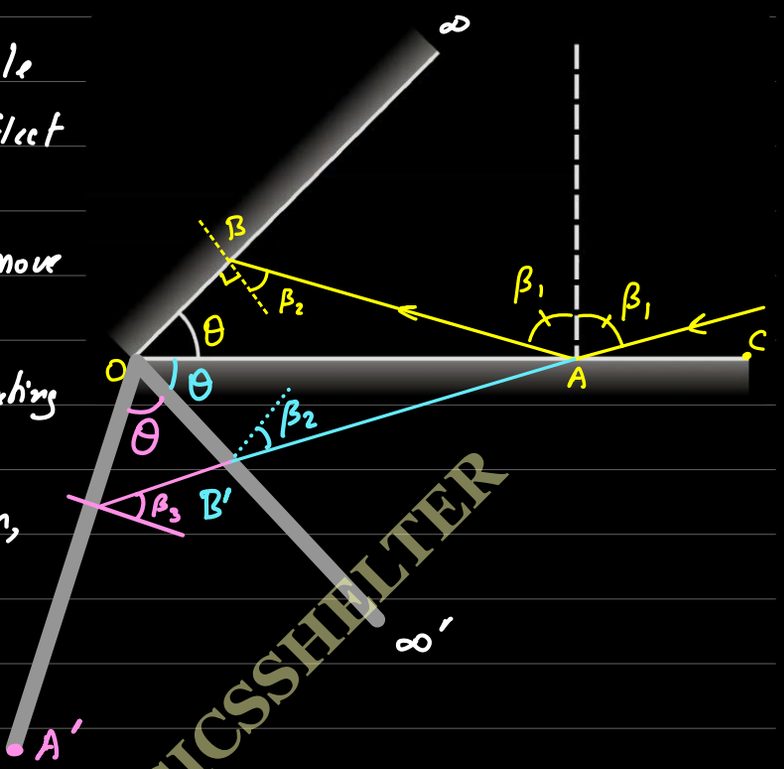
Solution: In a system of two mirrors joint at an edge, for finding successive angles of incidence, instead of reflecting the incident ray, we can reflect the next mirror in the first mirror and assume the ray to be moving straight. For instance in the adjacent figure, for find β_2 instead of reflecting the ray CA, we reflect the mirror $O\infty$ to form its image $O\infty'$ and let ray CA intersect it at B'. Now it can be easily seen that $\Delta OBA \cong OB'A$, hence,

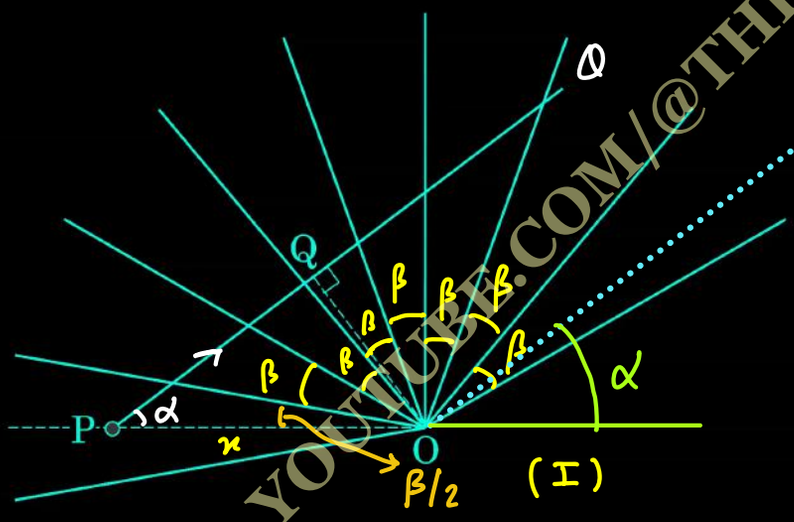


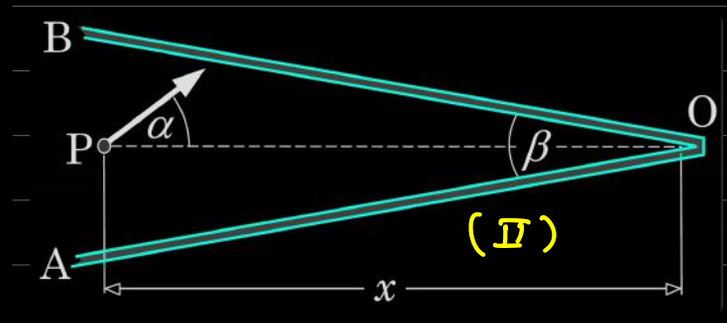
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the incidence angle on $O\infty'$ is also β_2. This process can be repeated again and again for getting successive angles of incidence.

For example, to find next angle of incidence β_3, we can now reflect the mirror OC in $O\infty'$ to form mirror OA' and still continue to move the ray straight. This pattern continues for purpose of calculating successive angles of incidence. By the ray diagram consideration, the cone problem is essentially same as the two mirror problem and we can keep flipping the cone successively to get the required incidence angles.







Based on our discussion above, we draw figure I to help us calculate successive angles of incidence. We draw a line $\overrightarrow{OT}$ parallel to ray $\overrightarrow{PQ}$, so long as the flipped cone edges are above $\overrightarrow{OT}$, there will be an intersection of PQ with some flipped edge. Thus to find number of reflections, we need to just count the number of edges in the angle POT. First edge occurs at angle $\beta/2$ from OP and after that they are spread within an angle of $\pi - \alpha - \beta/2$, with an angular separation β each.

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Now let the total number of reflections be n then: $\beta/2 + (n-1)\beta < \pi - \alpha$

$$\Rightarrow \quad (n-1) < \left(\pi - \alpha - \beta/2\right)\cdot\frac{1}{\beta} = \frac{2\pi - 2\alpha - \beta}{2\beta}$$

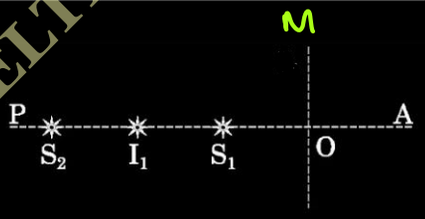
$$\Rightarrow \quad n < \frac{2\pi - 2\alpha + \beta}{2\beta} \Rightarrow n = \left[\frac{2\pi - 2\alpha + \beta}{2\beta}\right]$$

Putting in $\alpha = 37^\circ$, $\beta = 20^\circ$, $n = 7$ Ans.

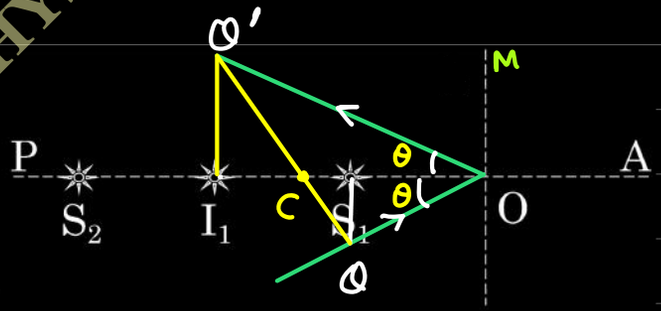
Now nearest distance is simply OP $\sin\alpha = n\sin\alpha = 6.0$ cm Ans.

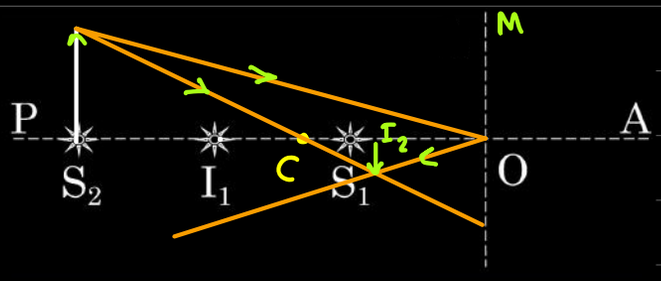
EXAMPLE:

In the arrangement shown, S_1 and S_2 are two point light sources placed on the optical axis PA of a Mirror of unknown nature and I_1 is the image of the source S_1. Distances S_2I_1, I_1S_1 and S_1O are equal. Draw a ray diagram to obtain image of the source S_2 and with the help of this diagram determine location of the image.



Solution: Since the object and the image are on same side of the mirror, the image must be invented. We imagine an elongated object S_1Q and form its elongated image I_1Q' by considering a ray reflected from pole. Now we locate C by joining QQ'. Once C is located, we can similarly find out location of image of S_2 as shown.





Analytical Solution: Let $OS_1 = S_1I_1 = x$, Now using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-x} + \frac{1}{-2x} = \frac{1}{f} \Rightarrow f = \frac{-2x}{3}$

Now for S_2: $\frac{1}{-3x} + \frac{1}{v} = \frac{-3}{2x} \Rightarrow \frac{1}{v} = \frac{-3}{2x} + \frac{1}{3x} = \frac{-9+2}{6x} = \frac{-7}{6x}$

$$\Rightarrow \quad v = \frac{-6x}{7}$$

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